

Install and manage WSUS for Update Management process-12

WSUS System Requirements

When you have decided to implement WSUS in your setup, you must first look into WSUS requirements.

The procedure to install Windows Server Update Services using Windows Internal Database (WID).

WSUS Firewall Ports / Exceptions

When you set up WSUS server, it is important that the server connects to Microsoft update to download updates. If there is a corporate firewall between WSUS and the Internet, you might have to configure that firewall to ensure WSUS can obtain updates.

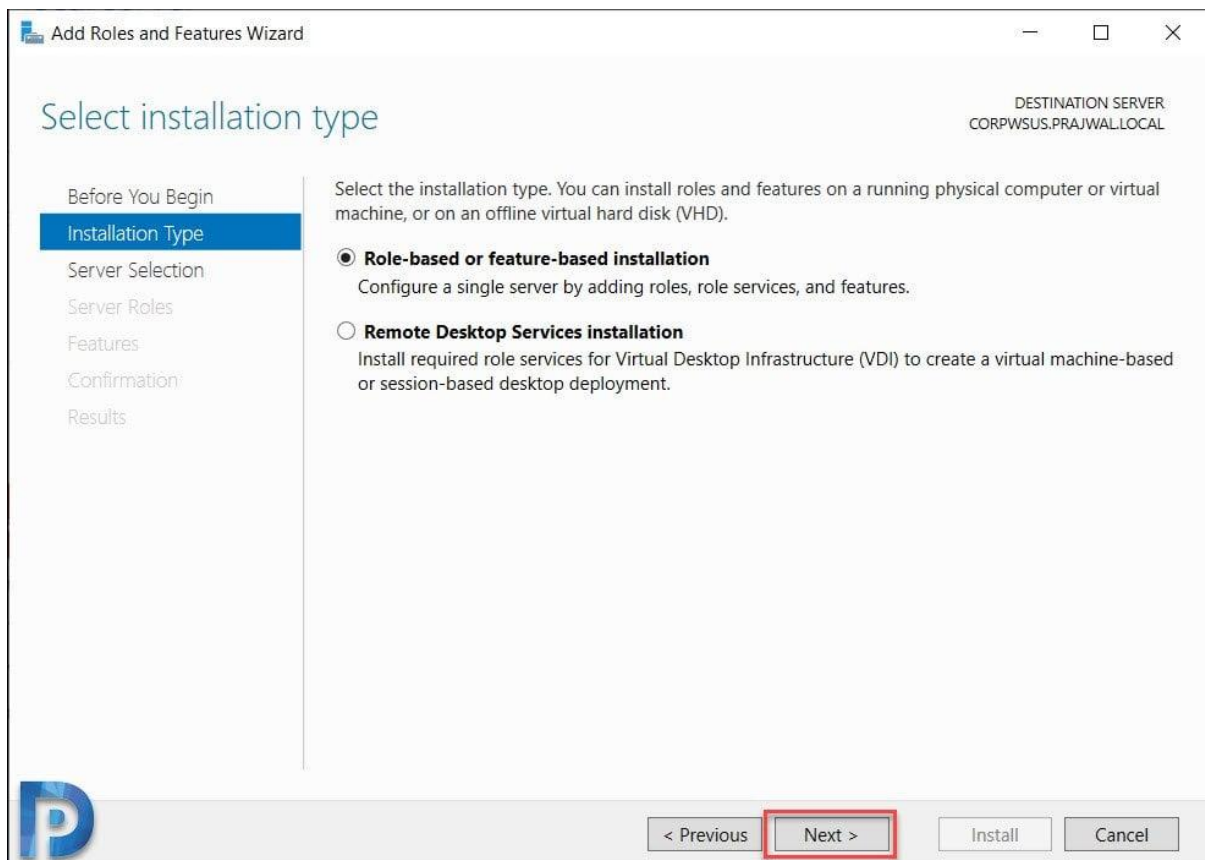
To obtain updates from Microsoft Update, the WSUS server uses port 443 for HTTPS protocol. You must allow Internet access from WSUS to the following list of URLs :-

- <http://windowsupdate.microsoft.com>
- http://*.windowsupdate.microsoft.com
- https://*.windowsupdate.microsoft.com
- http://*.update.microsoft.com
- https://*.update.microsoft.com
- http://*.windowsupdate.com
- <http://download.windowsupdate.com>
- <https://download.microsoft.com>
- http://*.download.windowsupdate.com
- <http://wustat.windows.com>
- <http://ntservicepack.microsoft.com>
- <http://go.microsoft.com>
- <http://dl.delivery.mp.microsoft.com>
- <https://dl.delivery.mp.microsoft.com>

Install WSUS Role on Windows Server 2019

The steps to install Windows Server Update Services (WSUS) Role on Windows Server 2019 include :-

- Log on to the Windows 2019 server on which you plan to install the WSUS server role using an account that is a member of the Local Administrators group.
- In Server Manager, click **Manage** and click add **Roles and Features**.
- On the Before you begin page, click **Next**.
- In the select installation type page, select **Role-based or feature-based installation** option. Click **Next**.



Select Role based or feature based installation

On the Server Selection page, verify the server name and click **Next**.

Add Roles and Features Wizard

Select destination server

DESTINATION SERVER
CORPWSUS.PRAJWALLOCAL

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool
☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
CORPWSUS.PRAJWALL...	192.168.100.101	Microsoft Windows Server 2019 Datacenter

1 Computer(s) found

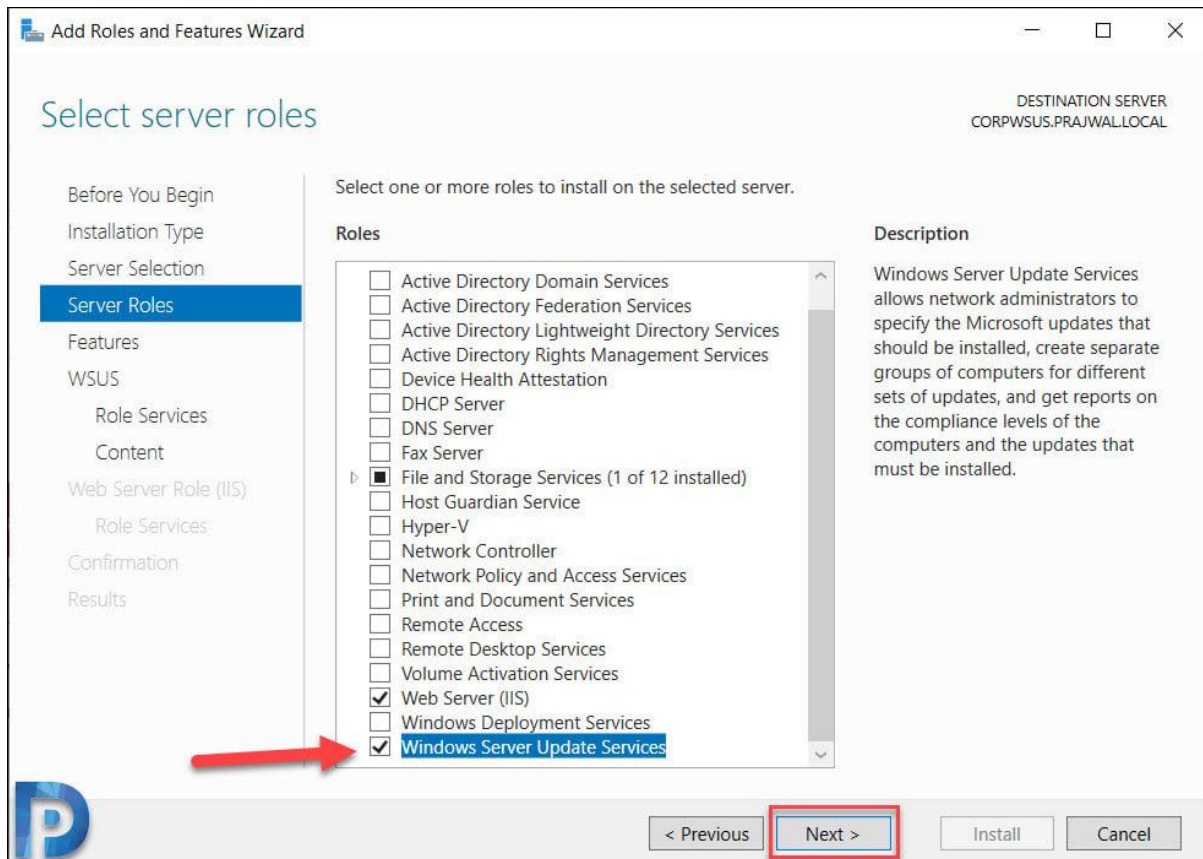
This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

 < Previous **Next >** Install Cancel

Select the server to install WSUS

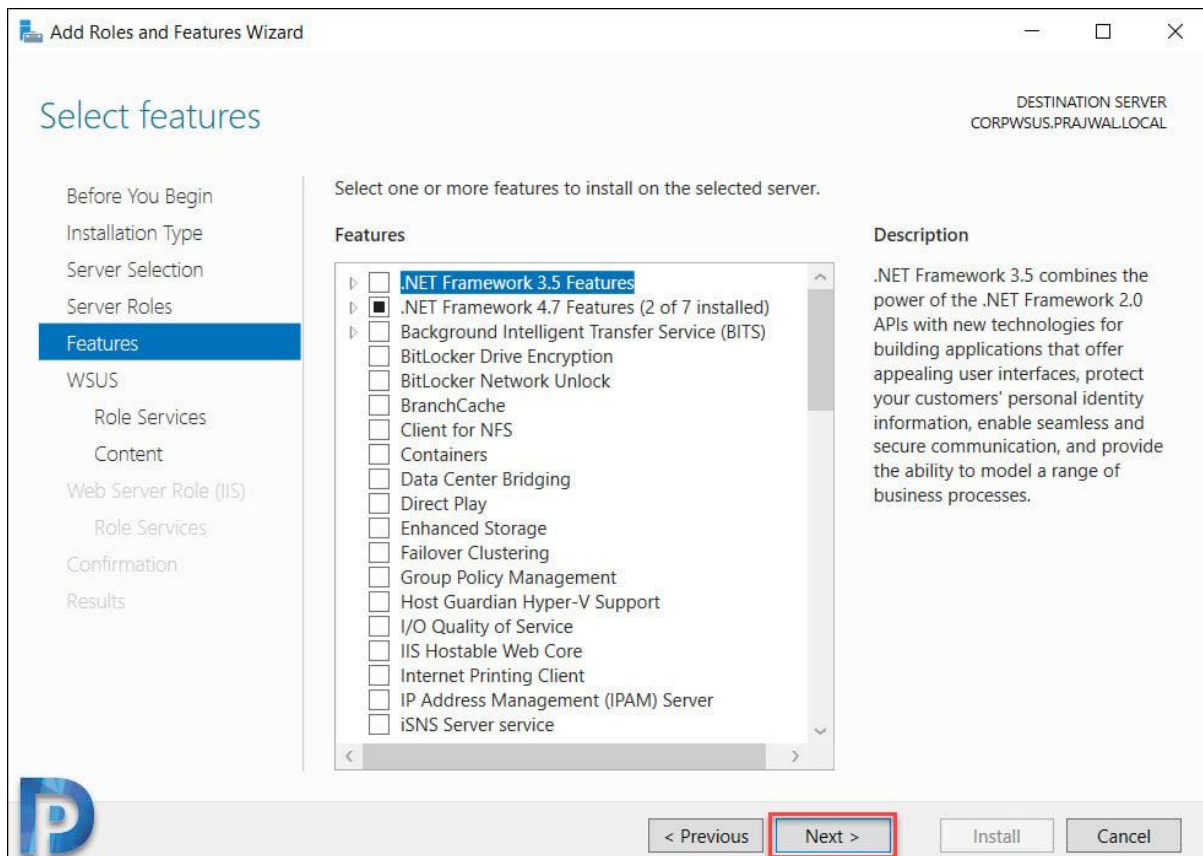
Server Roles – Windows Server Update Services

On the Server roles page, select the role “**Windows Server Update Services**“. You should see Add features that are required for Windows Server Update Services box. Click **Add Features**, and then click **Next**.

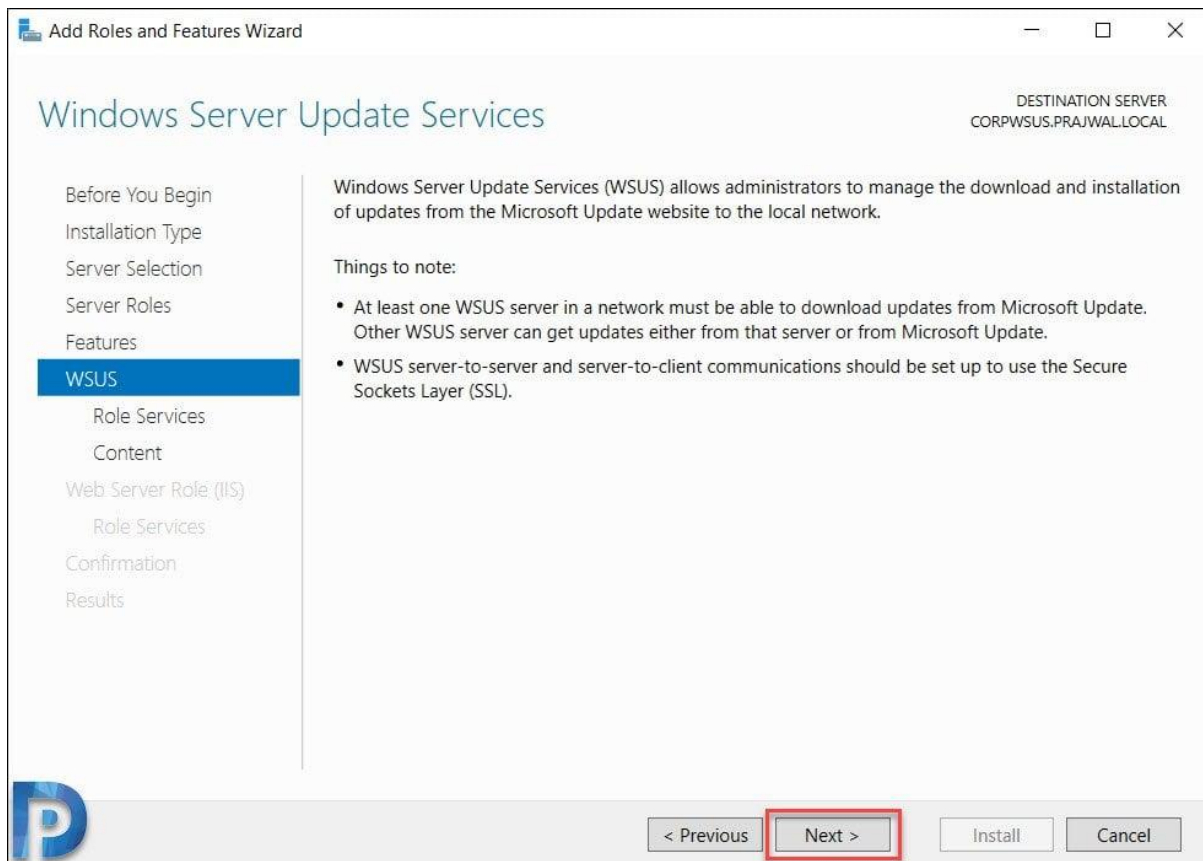


Select Windows Server Update Services role

On the Select features page, leave the options to default and click **Next**.

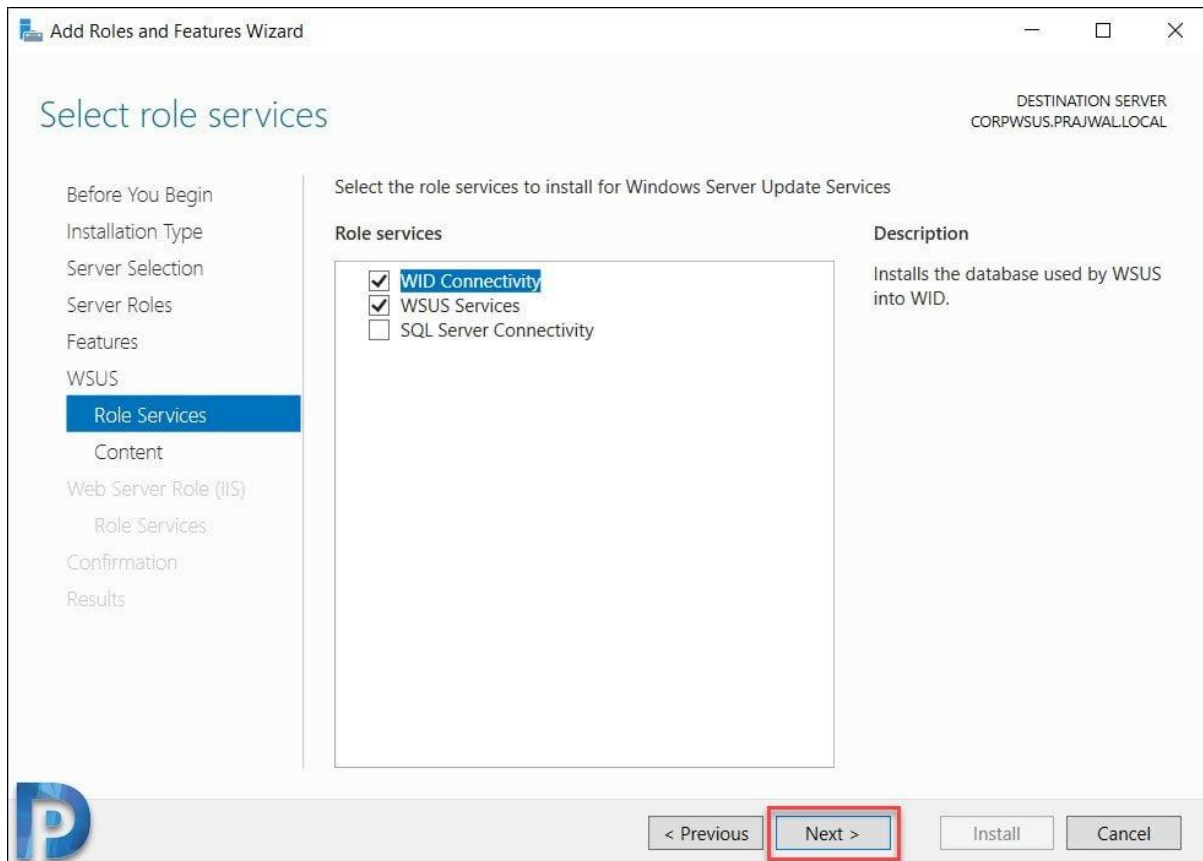


On the Windows Server Update Services page, click **Next**.



WSUS Database Type – Role Services

You must select role services / Database type to install for Windows Server Update services. Select **WID Connectivity** and **WSUS Services**. Click **Next**.



Select WID Connectivity and WSUS Services

WSUS Content Location

Specify a content location to store the updates. I would recommend storing the updates on another drive and not on your C: drive. The size of this folder can grow eventually and you don't want this folder to reside on C: drive. Hence choose either a separate drive or store the updates on remote server.

Click **Next**.

Add Roles and Features Wizard

Content location selection

DESTINATION SERVER
CORPWSUS.PRAJWAL.LOCAL

Before You Begin
Installation Type
Server Selection
Server Roles
Features
WSUS
Role Services
Content
Web Server Role (IIS)
Role Services
Confirmation
Results

If you have a drive formatted with NTFS and at least 6 GB of free disk space, you can use it to store updates for client computers to download quickly.

If you need to save disk space, clear the check box to store updates on Microsoft Update; downloads will be slower.

If you choose to store updates locally, updates are not downloaded to your WSUS server until you approve them. By default, when updates are approved, they are downloaded for all languages.

☒ Store updates in the following location (choose a valid local path on CORPWSUS.PRAJWAL.LOCAL, or a remote path):
E:\WSUS

< Previous **Next >** Install Cancel

WSUS Content Location

On the **Web Server Role (IIS)** page, click **Next**.

Add Roles and Features Wizard

Web Server Role (IIS)

DESTINATION SERVER
CORPWSUS.PRAJWAL.LOCAL

Before You Begin
Installation Type
Server Selection
Server Roles
Features
WSUS
Role Services
Content
Web Server Role (IIS)
Role Services
Confirmation
Results

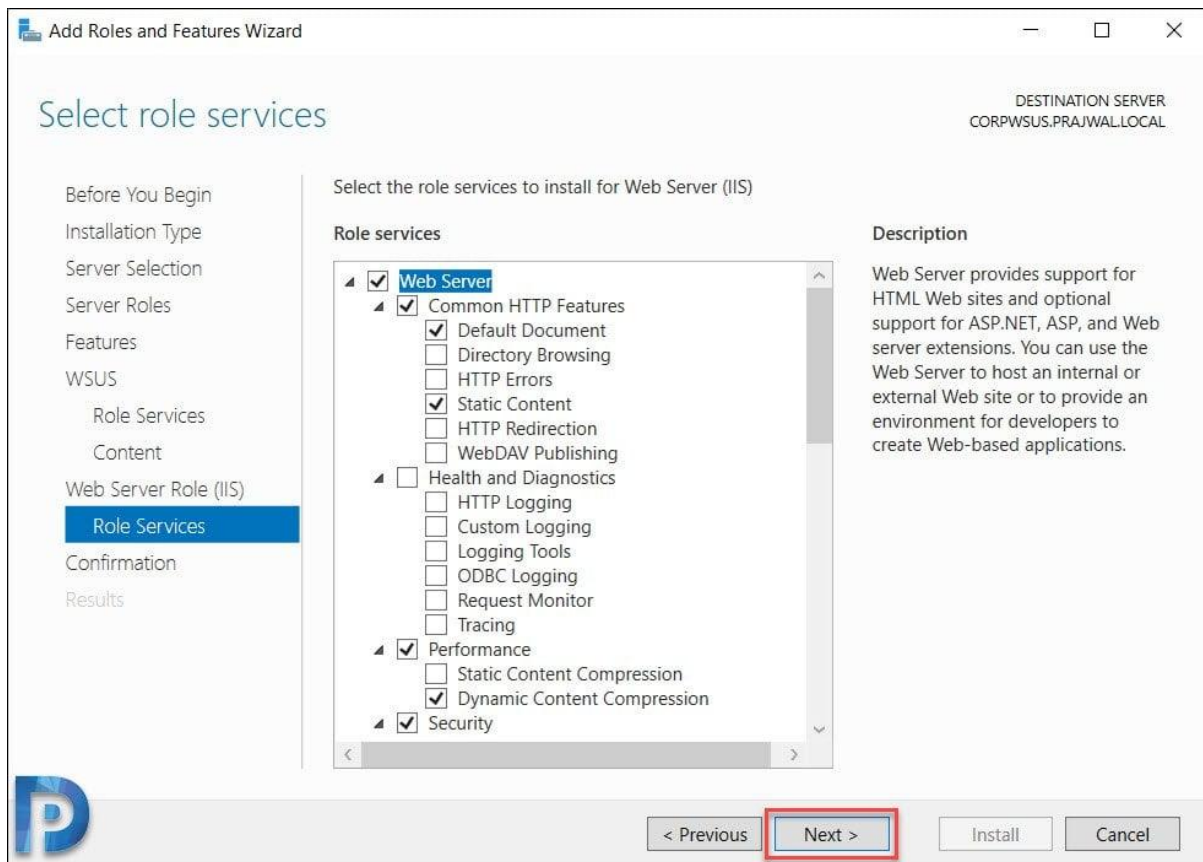
Web servers are computers that let you share information over the Internet, or through intranets and extranets. The Web Server role includes Internet Information Services (IIS) 10.0 with enhanced security, diagnostic and administration, a unified Web platform that integrates IIS 10.0, ASP.NET, and Windows Communication Foundation.

- The default installation for the Web Server (IIS) role includes the installation of role services that enable you to serve static content, make minor customizations (such as default documents and HTTP errors), monitor and log server activity, and configure static content compression.

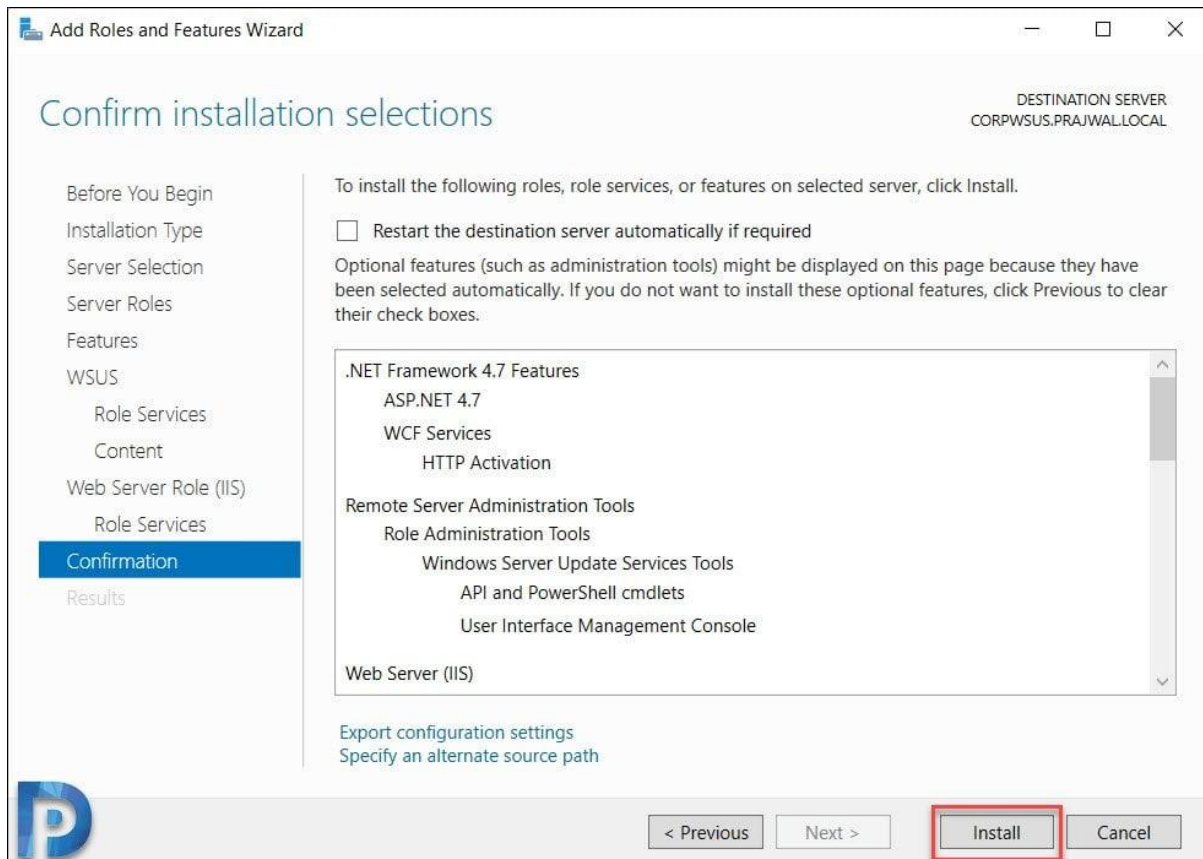
[More information about Web Server IIS](#)

< Previous **Next >** Install Cancel

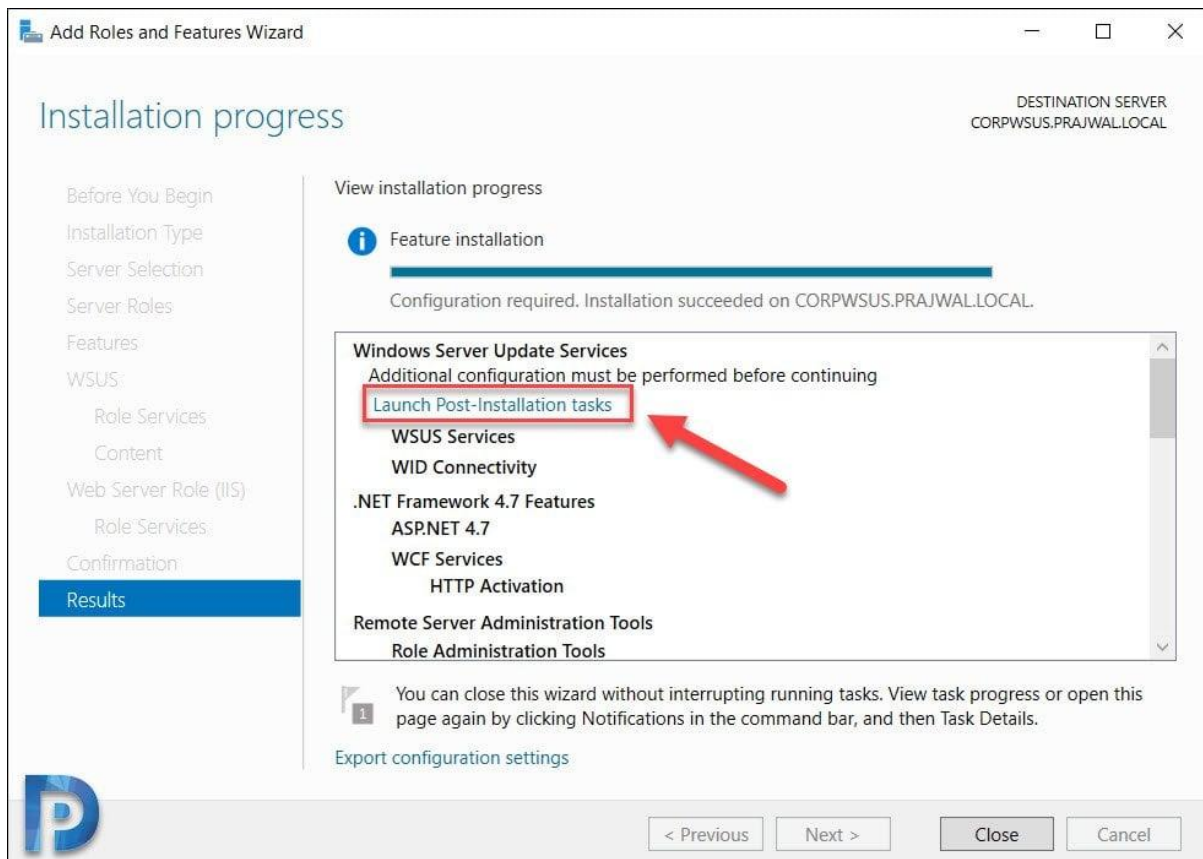
The role services to install web server (IIS) are select automatically. Do not change anything here and click **Next**.



A final confirmation before you install WSUS. Review the settings and click **Install**.

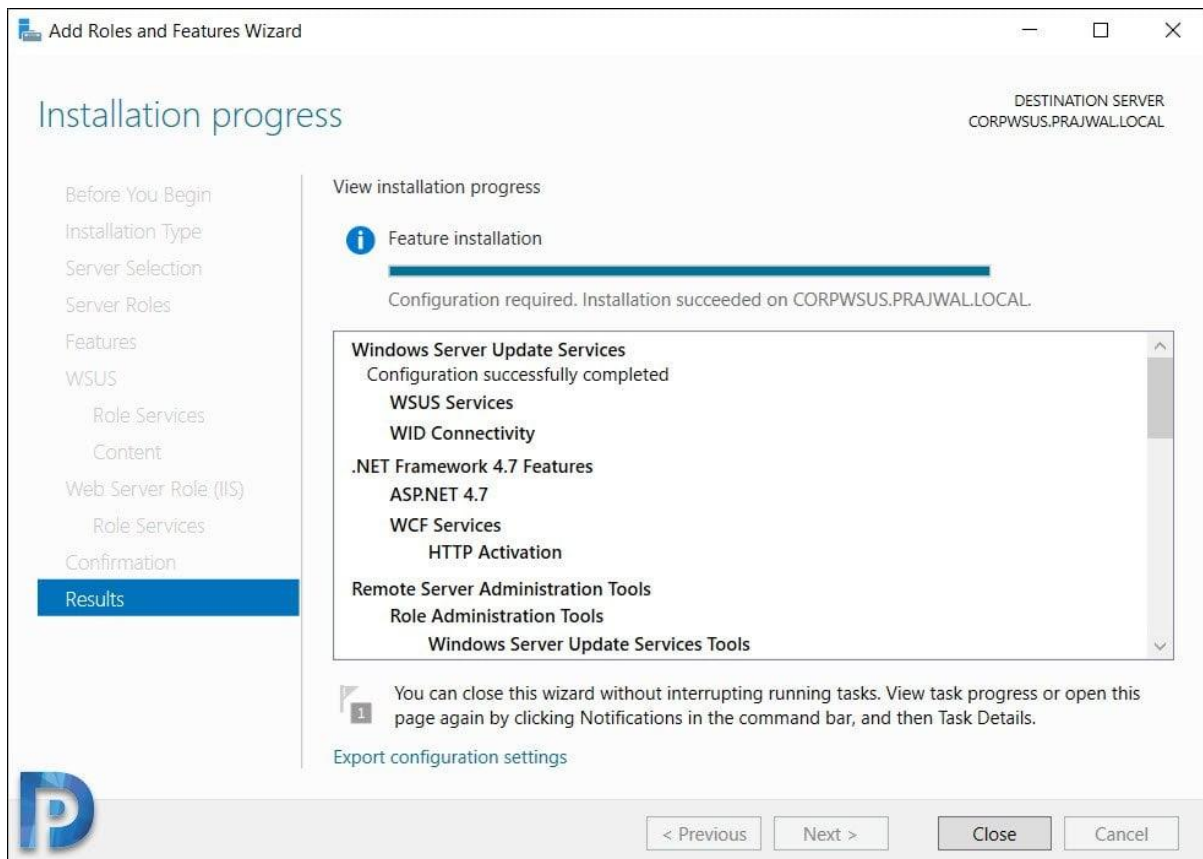


Once WSUS installation is complete, click **Launch Post-Installation tasks**.



Launch WSUS Post Installation tasks

Wait for the message Configuration successfully completed. Click **Close**.



WSUS configuration completed successfully

Configure Windows Server Update Services (WSUS)

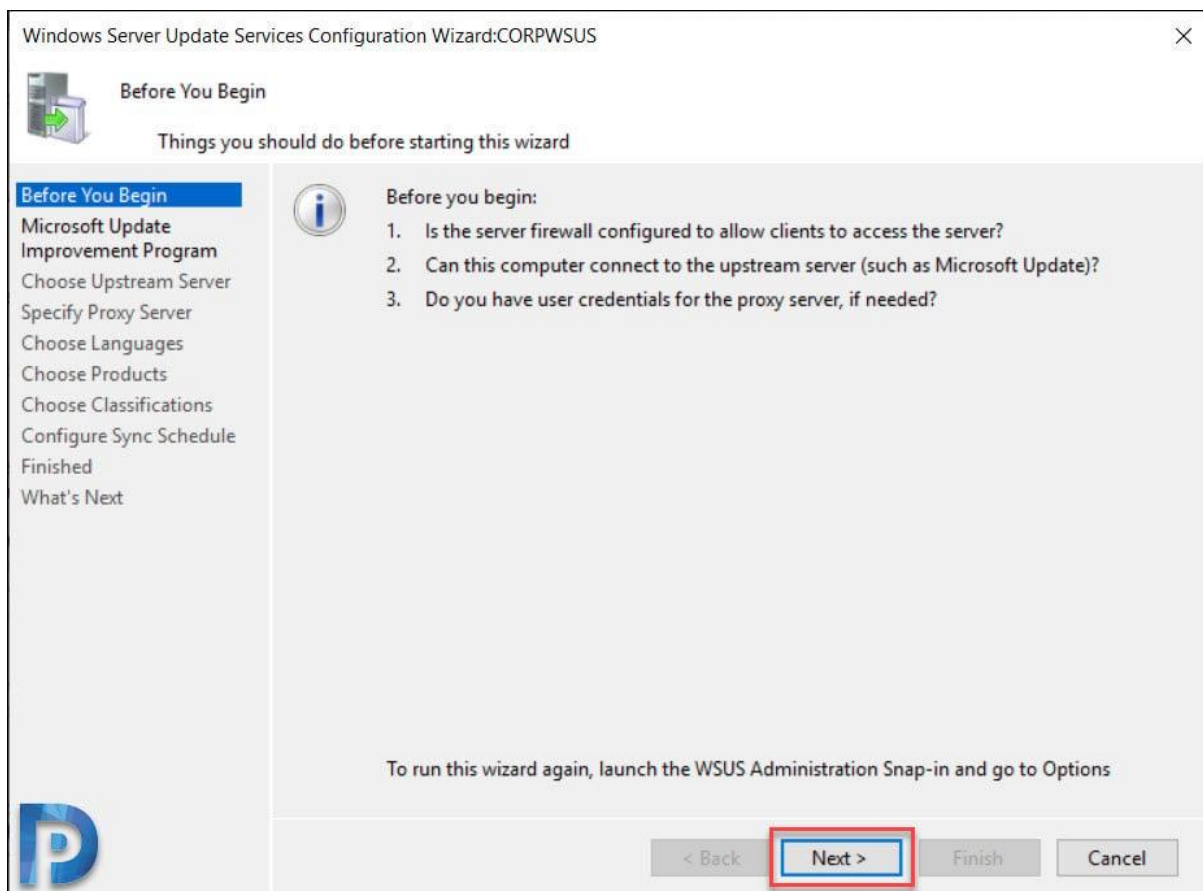
After you install WSUS, you can configure the WSUS server using WSUS Server configuration wizard. This is a one time configuration where you will configure some important WSUS options.

If you don't see a WSUS Server configuration wizard or if you have skipped it by mistake, don't worry. You can launch it by opening the **WSUS Console > Options > WSUS Server Configuration wizard**.

Note – Before you start to configure WSUS, some important points.

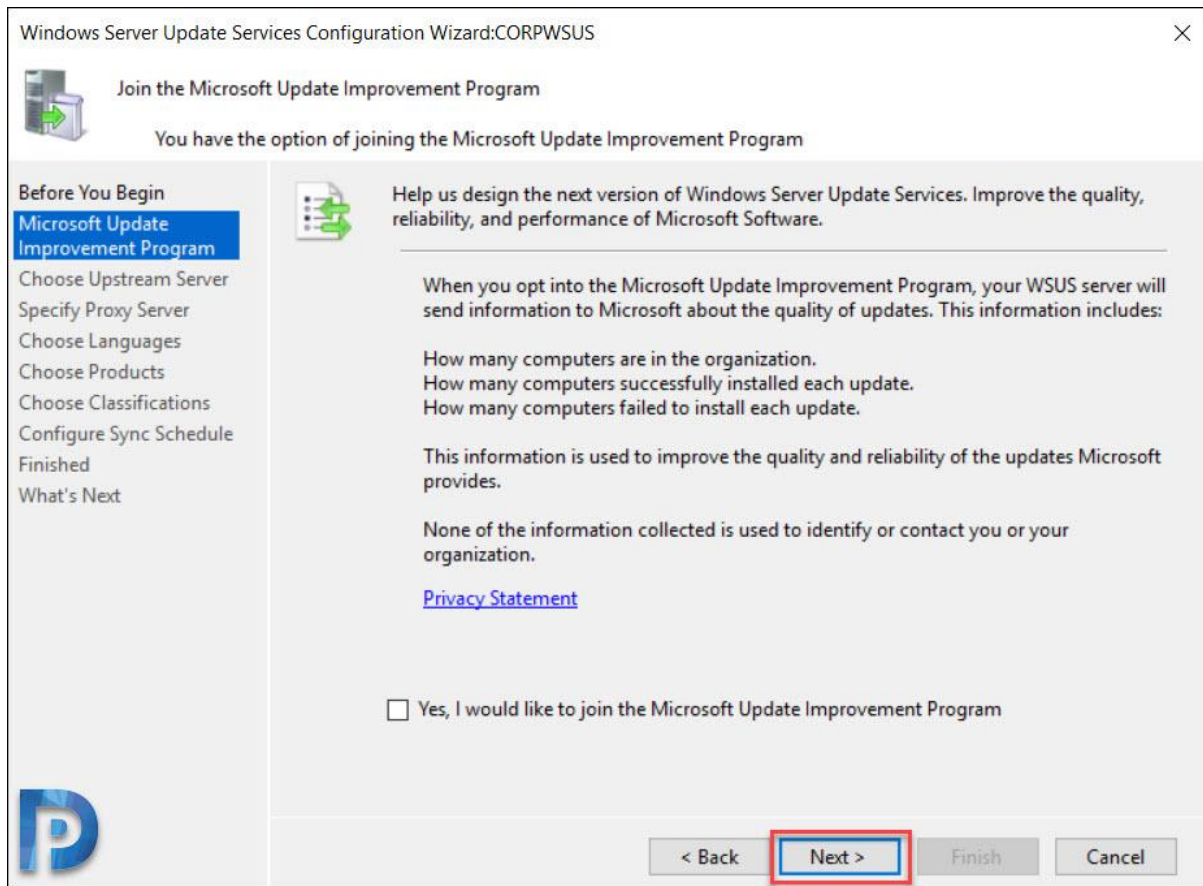
- Ensure the server firewall allows the clients to access the WSUS server. If the clients have issues connecting to WSUS server, updates won't be downloaded from server.
- The WSUS downloads the updates from upstream server which is Microsoft update in our case. So ensure the firewall allows the WSUS server to connect to Microsoft Update.
- In case there is a proxy server in your setup, you must enter the credentials for proxy server while configuring WSUS. Have them handy as they are required.

On the Before you begin page, click **Next**.



WSUS Configuration Wizard

Click **Next**.



Choose WSUS Upstream Server

This is an important section where you select the upstream server. You get two options.

- **Synchronize from Microsoft Update** – Selecting this option will download the updates from Microsoft update.
- **Synchronize from another Windows Server Update Services server** – Select this option if you want this WSUS server to download updates from already existing WSUS server. You must specify the server name and port number (8530) by default. If you are selecting the option to use SSL during updates synchronization, ensure that upstream WSUS server is also configured to support SSL.

Since this will be my only WSUS server, I will select **Synchronize from Microsoft Update**. Click **Next**.

Windows Server Update Services Configuration Wizard:CORPWSUS

Choose Upstream Server

Provide the upstream server from which you want to synchronize content

Before You Begin
Microsoft Update Improvement Program
Choose Upstream Server
Specify Proxy Server
Choose Languages
Choose Products
Choose Classifications
Configure Sync Schedule
Finished
What's Next

You can choose the upstream server from which your server synchronizes updates.

☒ Synchronize from Microsoft Update

☐ Synchronize from another Windows Server Update Services server

Server name:

Port number:

☐ Use SSL when synchronizing update information
If using SSL, ensure that the upstream WSUS server is also configured to support SSL.

☐ This is a replica of the upstream server
A replica server mirrors update approvals, settings, computers and groups from its parent. Updates can be approved only on the upstream server.

< Back **Next >** Finish Cancel

Choose Upstream Server

Proxy Server

Specify Proxy server information if you have got one. If this option is selected, ensure you specify proxy server name and port number. In addition to that specify the credentials to connect to the proxy server. If you want to enable basic authentication for the user connecting to the proxy server, click **Allow basic authentication (password in clear text)**.

Click **Next**.

Windows Server Update Services Configuration Wizard:CORPWSUS

 Specify Proxy Server

Provide proxy server settings for synchronizing updates with Microsoft Update

Before You Begin
Microsoft Update Improvement Program
Choose Upstream Server
Specify Proxy Server
Choose Languages
Choose Products
Choose Classifications
Configure Sync Schedule
Finished
What's Next

 If this server requires a proxy server to access the upstream server, you can configure the proxy server settings here.

☐ Use a proxy server when synchronizing

Proxy server name:

Port number:

☐ Use user credentials to connect to the proxy server

User name:

Domain:

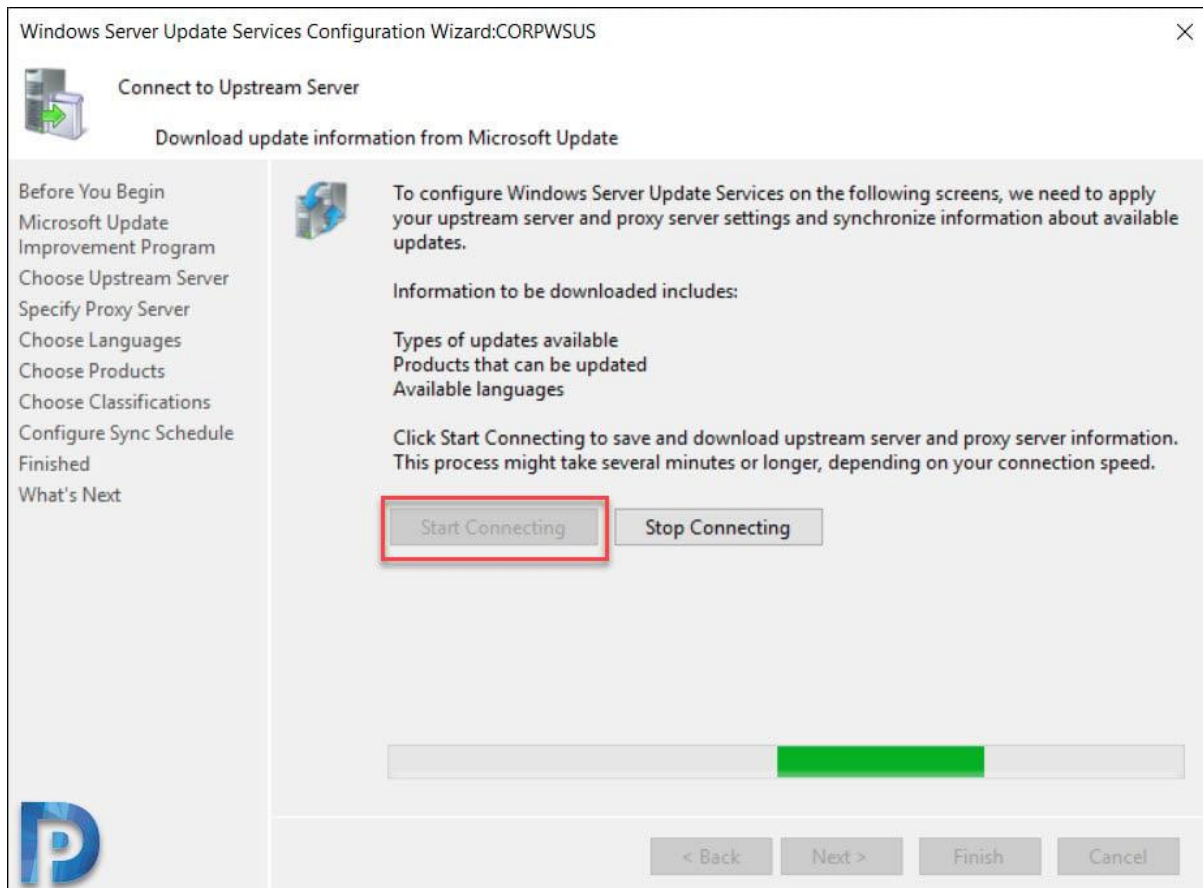
Password:

☐ Allow basic authentication (password is sent in cleartext)



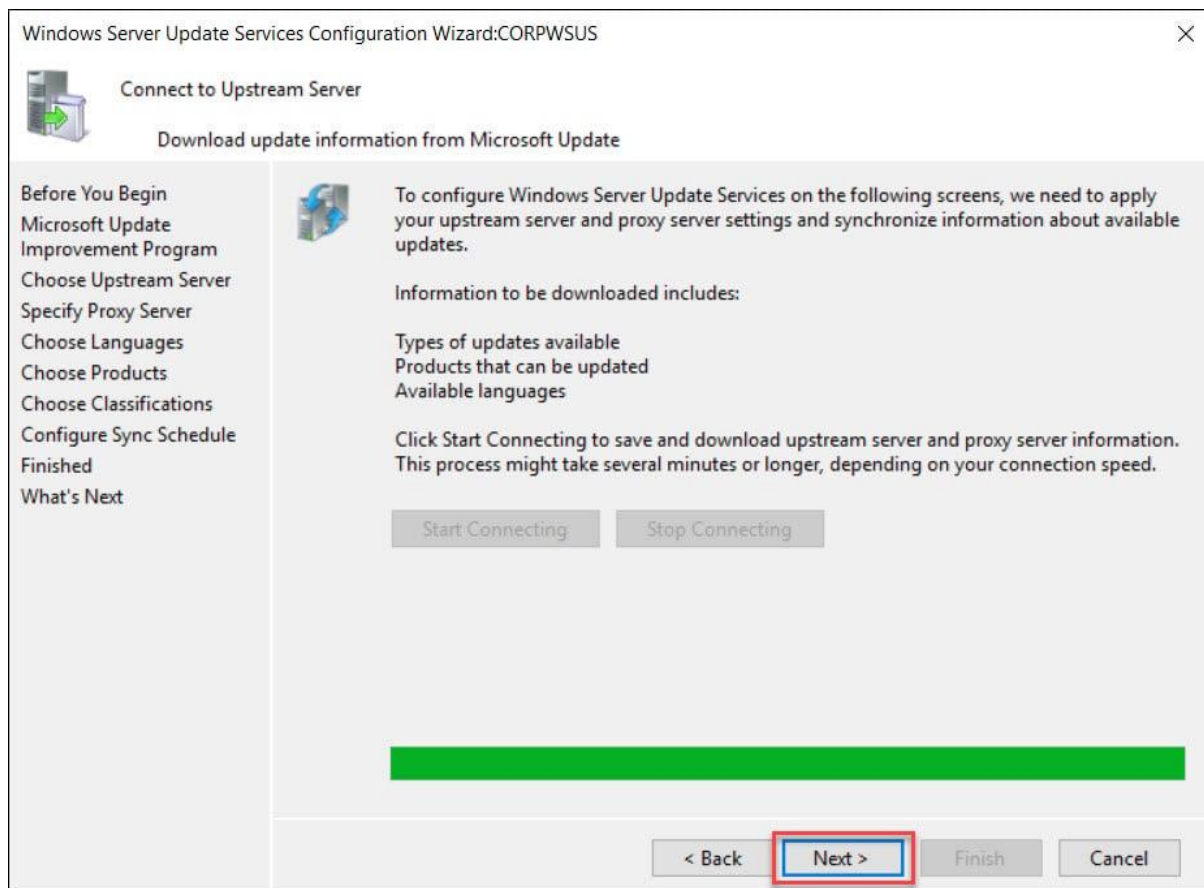
< Back **Next >** Finish Cancel

On the Connect to Upstream Server page, click **Start Connecting** button.



Download Update Information from Windows Update

Once it is complete, click **Next**.



Choose Languages for Updates

On the Choose Languages page, you have the option to select the languages from updates. If you choose to download updates in all languages, you would find updates with all languages in the WSUS console.

However if you choose to get updates only for specific languages, select **Download updates only in these languages**. Select the languages for which you want updates.

Click **Next**.

Windows Server Update Services Configuration Wizard: CORPWSUS

Choose Languages

Select the languages for which this server will download updates

Before You Begin
 Microsoft Update Improvement Program
 Choose Upstream Server
 Specify Proxy Server
Choose Languages
 Choose Products
 Choose Classifications
 Configure Sync Schedule
 Finished
 What's Next

If you are storing update files locally, you can filter the updates downloaded to your server by language. Choosing individual languages will affect which computers can be updated on this server and any downstream servers.

☐ Download updates in all languages, including new languages

☒ Download updates only in these languages:

☐ Arabic	☐ French	☐ Norwegian
☐ Bulgarian	☐ German	☐ Polish
☐ Chinese (Hong Kong S.A.R.)	☐ Greek	☐ Portuguese
☐ Chinese (Simplified)	☐ Hebrew	☐ Portuguese (E)
☐ Chinese (Traditional)	☐ Hindi	☐ Romanian
☐ Croatian	☐ Hungarian	☐ Russian
☐ Czech	☐ Italian	☐ Serbian (Latin)
☐ Danish	☐ Japanese	☐ Slovak
☐ Dutch	☐ Japanese (NEC)	☐ Slovenian
☒ English	☐ Korean	☐ Spanish
☐ Estonian	☐ Latvian	☐ Swedish
☐ Finnish	☐ Lithuanian	☐ Thai

< Next > Finish Cancel

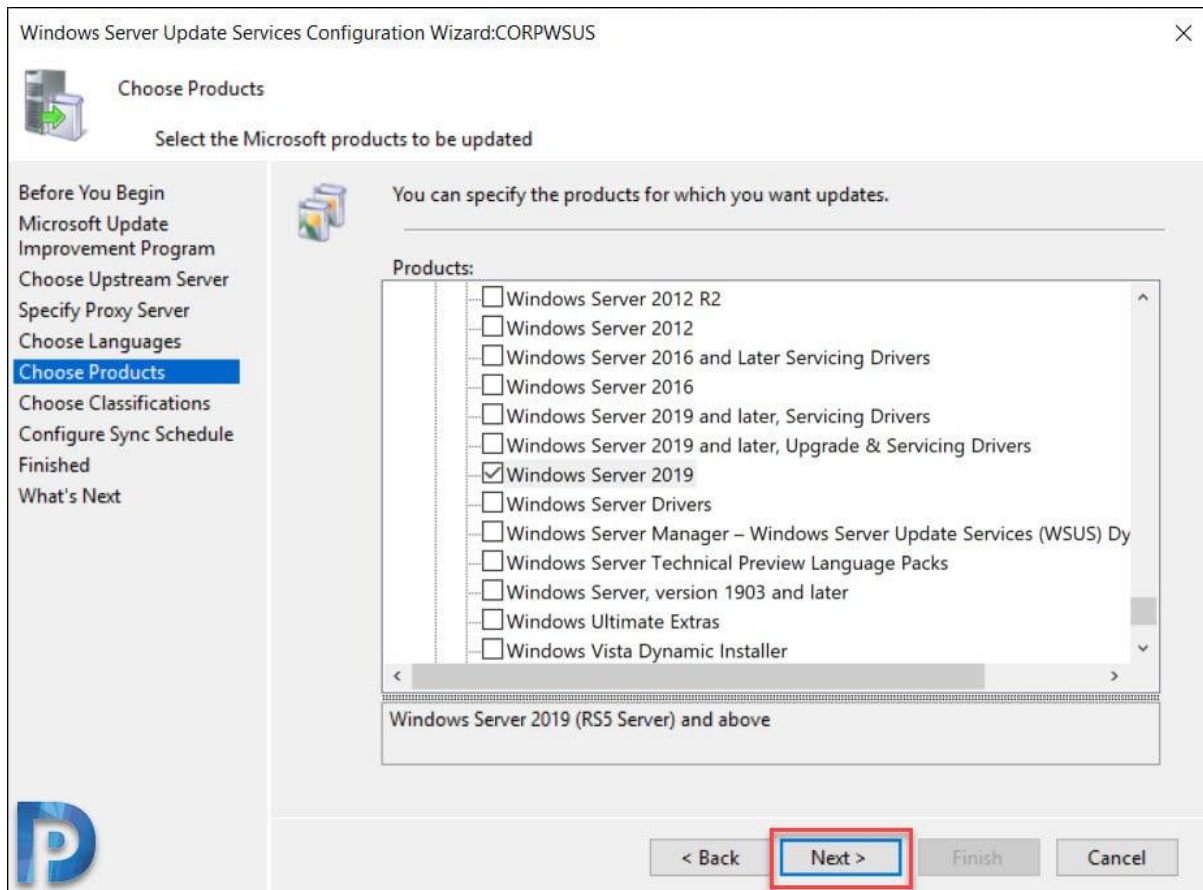
Windows Updates Languages

Choose Products

This is the page where you select the products for which you want the updates. A product is a specific edition of an operating system or application.

From the list of products you can select individual products or product families for which you want your server to synchronize updates. In this case I am going to select **Windows Server 2019** and **Windows 10 1903** as products.

Click **Next**.

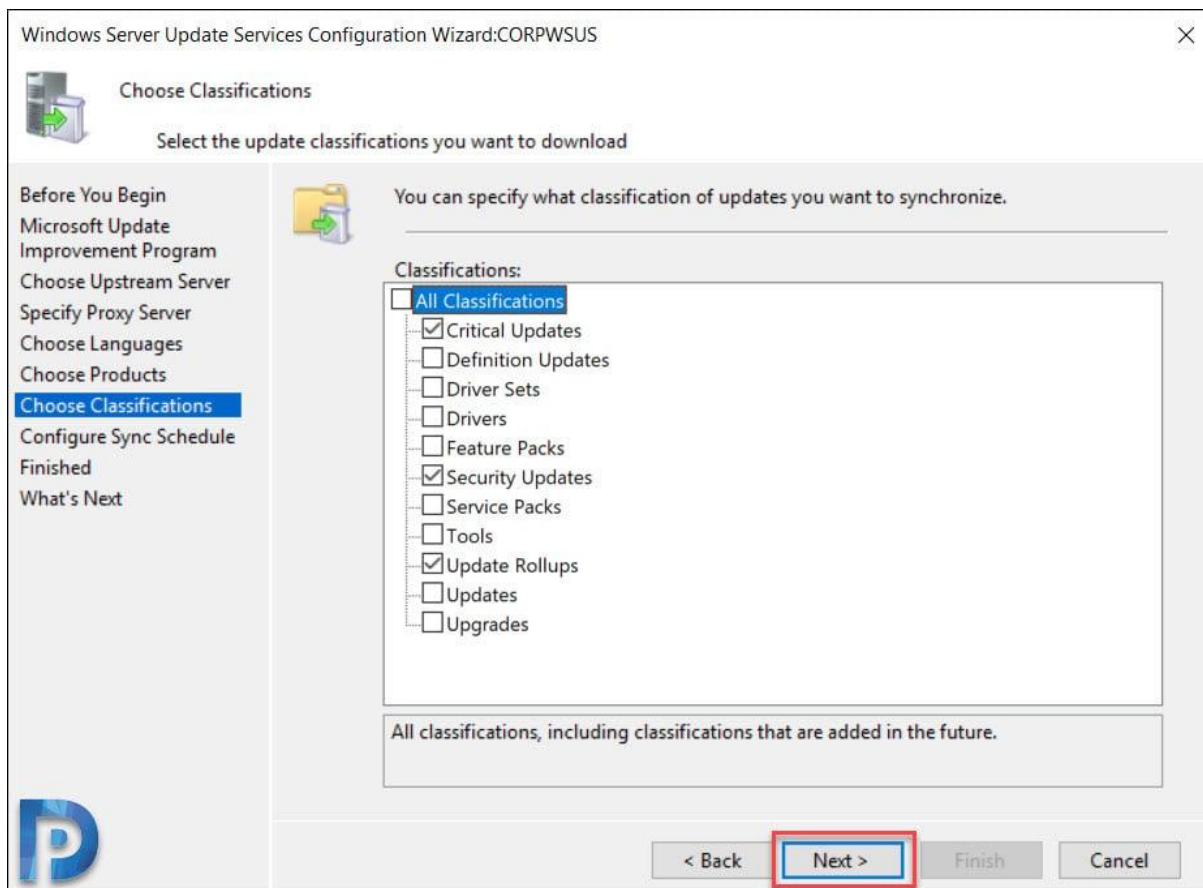


Select the Microsoft Products

Choose Update Classifications

In the beginning of the post I have listed the types of updates. On the Choose Classifications page, select the required classifications. I have selected **Critical Updates**, **Security Updates** and **Update Rollups**.

Click **Next**.



WSUS Update Classifications

Configure WSUS Synchronization Schedule

You must decide on how do you want to perform WSUS sync. The **Set Sync Schedule** page lets you select whether to perform synchronization manually or automatically.

If you choose **Synchronize manually**, you must manually start the synchronization process from the WSUS Administration Console. With this option selected, you have to manually perform the sync every time. Therefore do not select this option if you are setting up the WSUS in production.

If you choose **Synchronize automatically**, the WSUS server will synchronize at set intervals. You can set the time of First synchronization. Then set the number of synchronizations per day. From the drop-down you can choose the value between 1-24.

Click **Next**.

Windows Server Update Services Configuration Wizard: CORPWSUS

 Set Sync Schedule

Configure when this server synchronizes with Microsoft Update

Before You Begin
Microsoft Update Improvement Program
Choose Upstream Server
Specify Proxy Server
Choose Languages
Choose Products
Choose Classifications
Configure Sync Schedule
Finished
What's Next

 You can synchronize updates manually or set a schedule for daily automatic synchronization.

☒ Synchronize manually

☐ Synchronize automatically

First synchronization: 11:34:08 PM

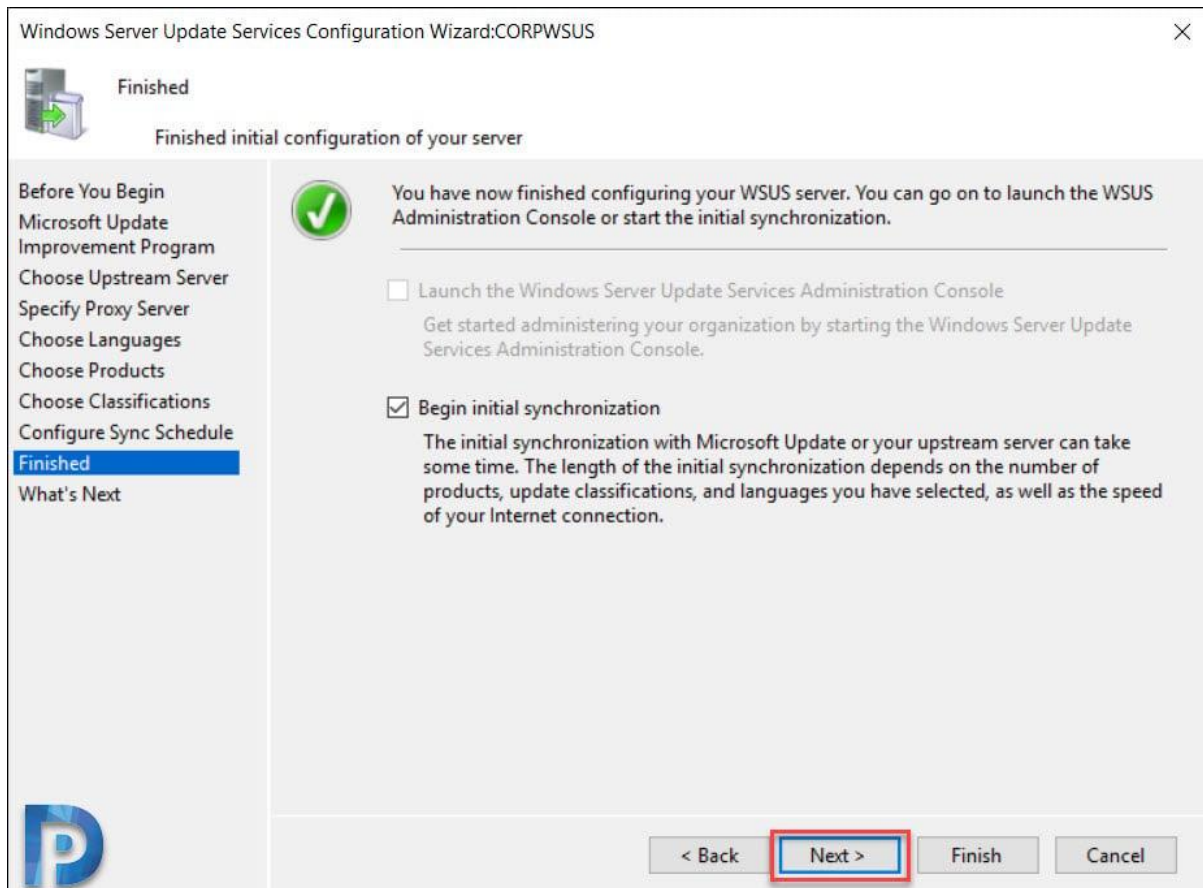
Synchronizations per day: 1

Note that when scheduling a daily synchronization from Microsoft Update, the synchronization start time will have a random offset up to 30 minutes after the specified time.

 < Back **Next >** Finish Cancel

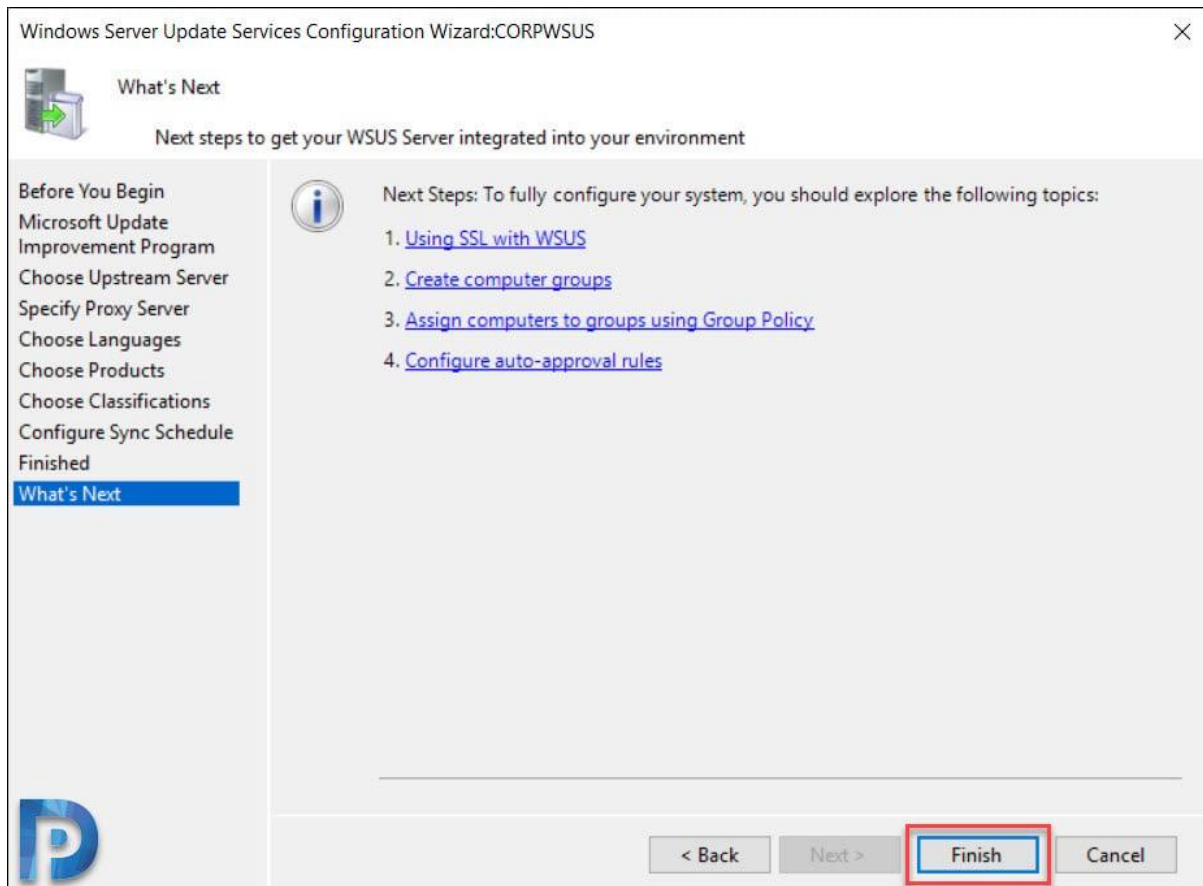
Configure WSUS Synchronization Schedule

Click **Begin initial synchronization**. Click **Next**.



Begin WSUS Initial synchronization

Finally on the last page, click **Finish**. This completes the steps to configure WSUS.



Configure Group Policy Settings for WSUS

After you install and configure WSUS, the next important task is to configure group policy settings for automatic updates. The new clients still don't know about the new WSUS server that you just setup. Using group policy you can point your client machines to new WSUS server.

In an active directory environment, you can use Group Policy specify the WSUS server. The group policy settings will be used to obtain automatic updates from Windows Server Update Services (WSUS).

You can create the group policy and apply it at domain level. Or you can create and apply the GPO to a specific OU (containing your computers).

While there are many Windows Update policy settings, I am going to configure few of them. For a list of all windows update policy settings, read this [article](#) from Microsoft.

Configure Automatic Updates WSUS

To configure Automatic Updates group policy settings for WSUS

- Open the Group Policy Management console, and open an existing GPO or create a new one.

- Navigate to **Computer Configuration > Policies > Administrative Templates > Windows Components > Windows Update**.
- Double-click **Configure Automatic Updates** and set it to **Enabled**.

Under Configure automatic updating, select the desired option. Under **Schedule install day**, select the day when you want the updates to be installed. Set the scheduled install time.

Configure Automatic Updates

Previous Setting Next Setting

☐ Not Configured Comment:
☒ **Enabled**
☐ Disabled

Supported on: Windows XP Professional Service Pack 1 or At least Windows 2000 Service Pack 3

Options: Help:

Configure automatic updating: 1
 3 - Auto download and notify for install

The following settings are only required and applicable if 4 is selected.

☐ Install during automatic maintenance

Scheduled install day: 2
 0 - Every day

Scheduled install time: 17:00

If you have selected "4 - Auto download and schedule the install" for your scheduled install day and specified a schedule, you also have the option to limit updating to a weekly, bi-weekly or monthly occurrence, using the options below:

Help: Specifies whether this computer will receive security updates and other important downloads through the Windows automatic updating service.

Note: This policy does not apply to Windows RT.

This setting lets you specify whether automatic updates are enabled on this computer. If the service is enabled, you must select one of the four options in the Group Policy Setting:

2 = Notify before downloading and installing any updates.

When Windows finds updates that apply to this computer, users will be notified that updates are ready to be downloaded. After going to Windows Update, users can download and install any available updates.

3 = (Default setting) Download the updates automatically and notify when they are ready to be installed

OK Cancel Apply

Configure WSUS Automatic Updates

In case you select Auto download and schedule the updates install, you get some options to limit updating frequency. If you have configured the settings, click **Apply** and **OK**.

Configure Automatic Updates

Previous Setting Next Setting

☐ Not Configured Comment:
☒ Enabled
☐ Disabled

Supported on: Windows XP Professional Service Pack 1 or At least Windows 2000 Service Pack 3

Options:

Scheduled install time: 17:00

If you have selected "4 – Auto download and schedule the install" for your scheduled install day and specified a schedule, you also have the option to limit updating to a weekly, bi-weekly or monthly occurrence, using the options below:

☒ Every week
☐ First week of the month
☐ Second week of the month
☐ Third week of the month
☐ Fourth week of the month
☐ Install updates for other Microsoft products

Help:

Specifies whether this computer will receive security updates and other important downloads through the Windows automatic updating service.

Note: This policy does not apply to Windows RT.

This setting lets you specify whether automatic updates are enabled on this computer. If the service is enabled, you must select one of the four options in the Group Policy Setting:

2 = Notify before downloading and installing any updates.

When Windows finds updates that apply to this computer, users will be notified that updates are ready to be downloaded. After going to Windows Update, users can download and install any available updates.

3 = (Default setting) Download the updates automatically and notify when they are ready to be installed

OK Cancel Apply

Configure WSUS Automatic Updates

Specify Intranet Microsoft Update Service Location

The next setting that you should configure is specify an intranet Microsoft update service location. The idea behind this is to ensure the client computers contact the specified intranet server instead of downloading updates from internet. Unless you configure this policy setting, the client computers wouldn't know about the intranet server.

To enable the policy, click **Enabled**. Specify the intranet update service and intranet statistics server. Click **Apply** and **OK**.

Specify intranet Microsoft update service location

Specify intranet Microsoft update service location

Previous Setting Next Setting

☐ Not Configured
 ☒ Enabled
 ☐ Disabled

Comment:

Supported on: At least Windows XP Professional Service Pack 1 or Windows 2000 Service Pack 3, excluding Windows RT

Options:

Set the intranet update service for detecting updates:

http://corpwsus.prajwal.local:8530

Set the intranet statistics server:

http://corpwsus.prajwal.local:8530

Set the alternate download server:

(example: http://IntranetUpd01)

☐ Download files with no Url in the metadata if alternate download server is set.

Help:

Specifies an intranet server to host updates from Microsoft Update. You can then use this update service to automatically update computers on your network.

This setting lets you specify a server on your network to function as an internal update service. The Automatic Updates client will search this service for updates that apply to the computers on your network.

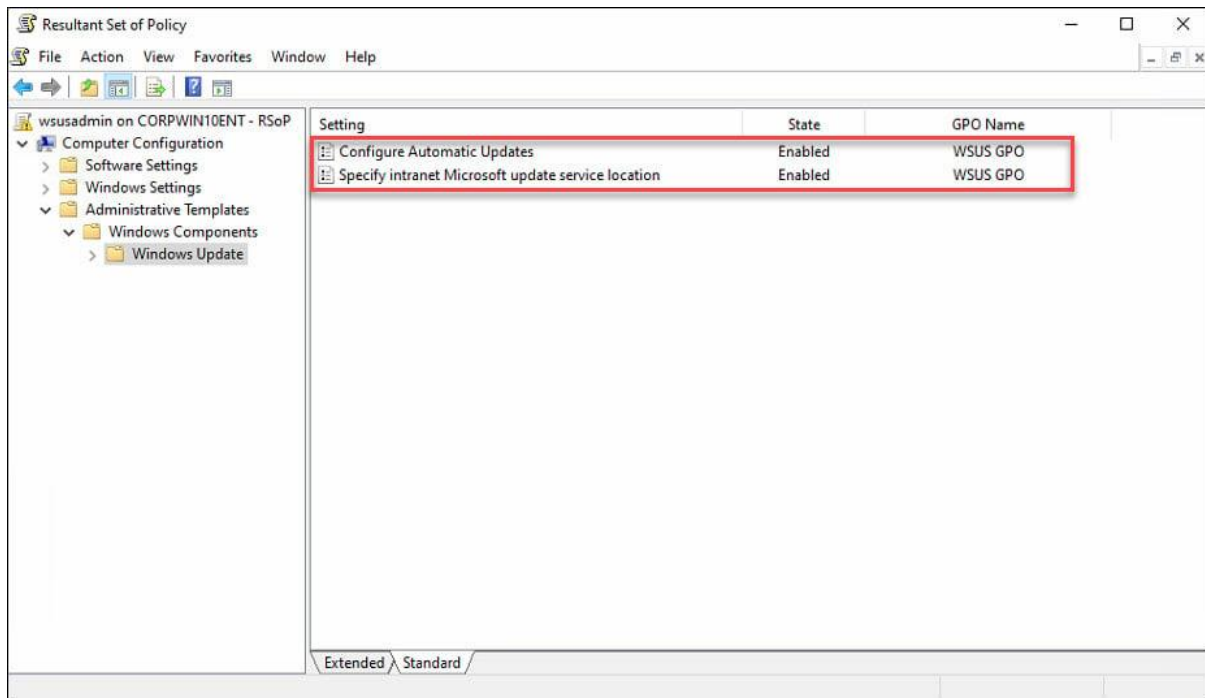
To use this setting, you must set two server name values: the server from which the Automatic Updates client detects and downloads updates, and the server to which updated workstations upload statistics. You can set both values to be the same server. An optional server name value can be specified to configure Windows Update Agent to download updates from an alternate download server instead of the intranet update service.

If the status is set to Enabled, the Automatic Updates client connects to the specified intranet Microsoft update service (or

OK Cancel Apply

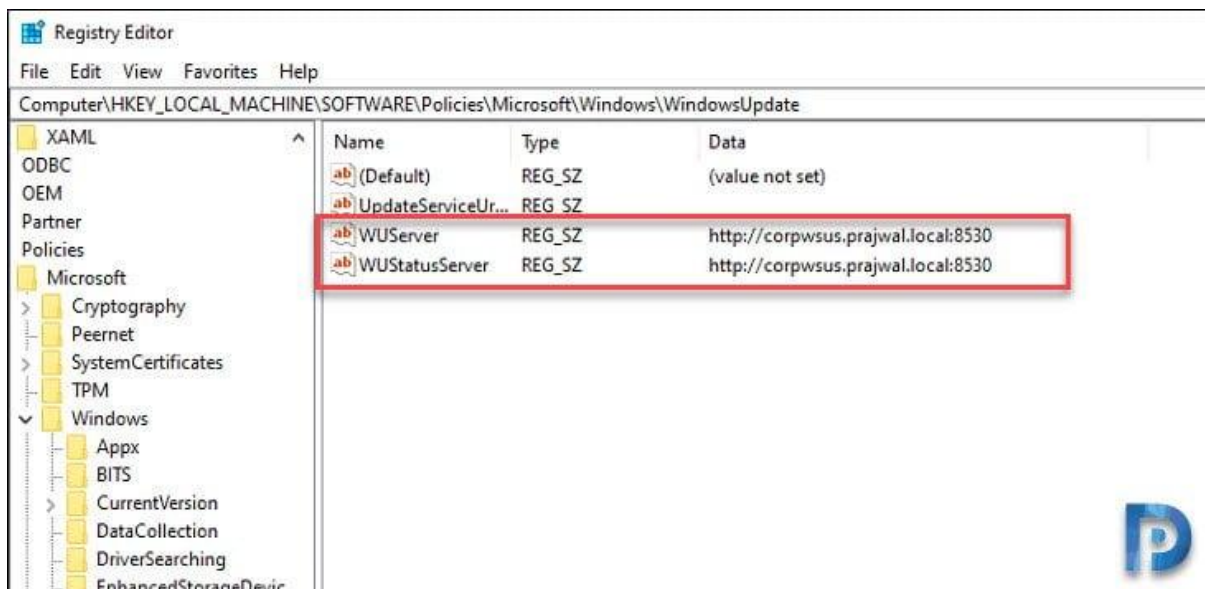
Specify intranet Microsoft Update service location

On the client computer, check the resultant set of policy to confirm if the WSUS GPO is applied.



You can also verify the intranet update service location on client computers using registry. On the client computer, open Registry Editor and go to **HKLM\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate**.

Check the values of **WUServer** and **WUStatusServer** and confirm if the values match the one that you supplied in WSUS GPO.



Verify intranet Microsoft Update service location

Configure WSUS computer groups

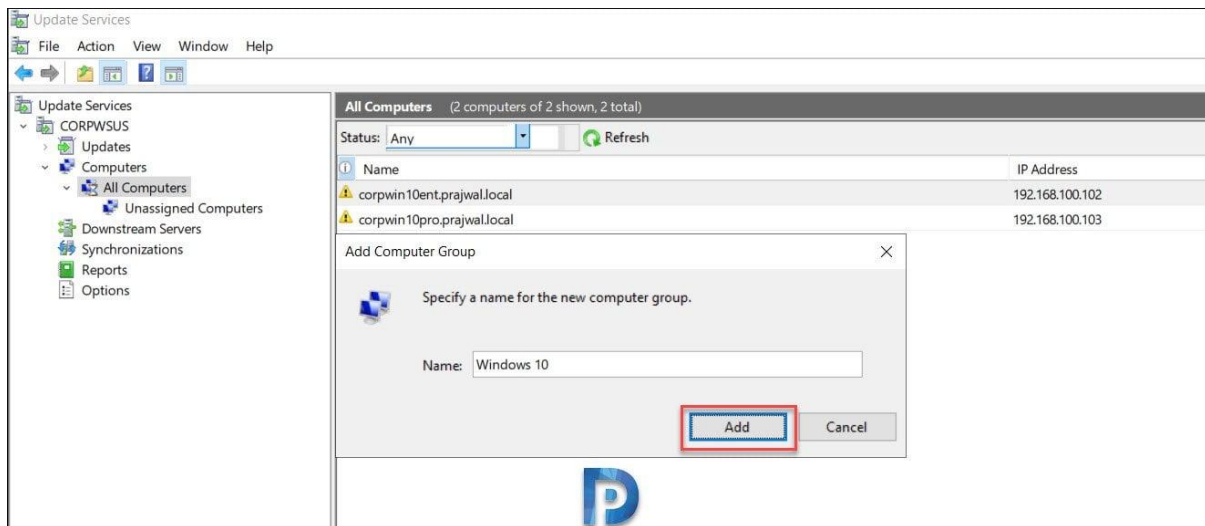
By creating computer groups you can first test and target updates to specific computers. When you open WSUS console, you will find two default computer groups – **All computers** and **Unassigned computers**.

You can create custom computer groups to manage updates in your organization. As per Microsoft you must create at least one computer group in the WSUS console. Test updates before you deploy them to other computers in your organization.

To create a new computer group in WSUS console

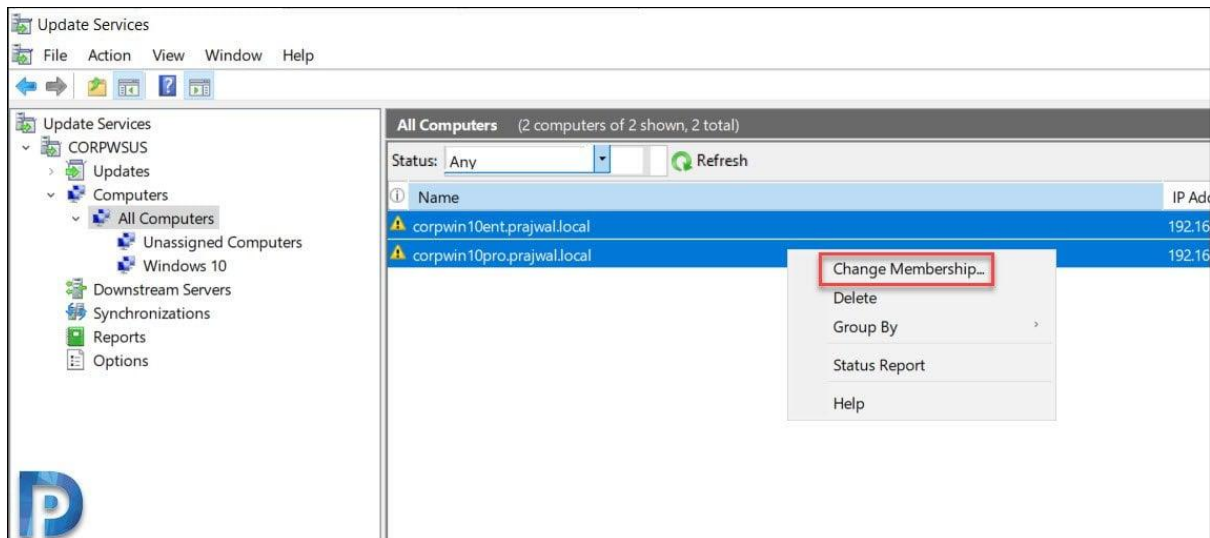
In the WSUS Administration Console, under Update Services, expand the WSUS server. Expand computers, right-click All computers, and then click **Add computer Group**.

In the add computer Group dialog box, specify the name of the new group, and then click **Add**.



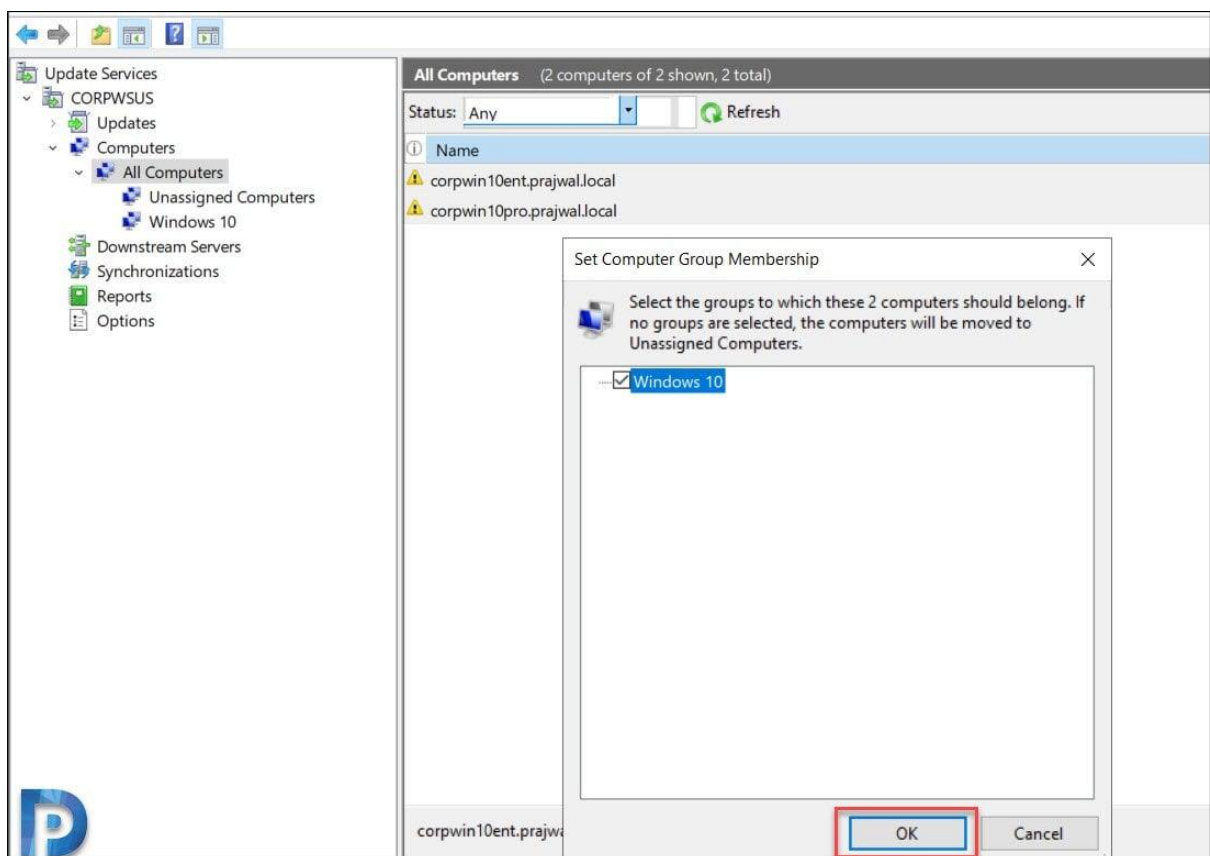
Add new computer group

Click **All Computers** and you should see list of computers. Select the computers, right click and click **Change Membership**.



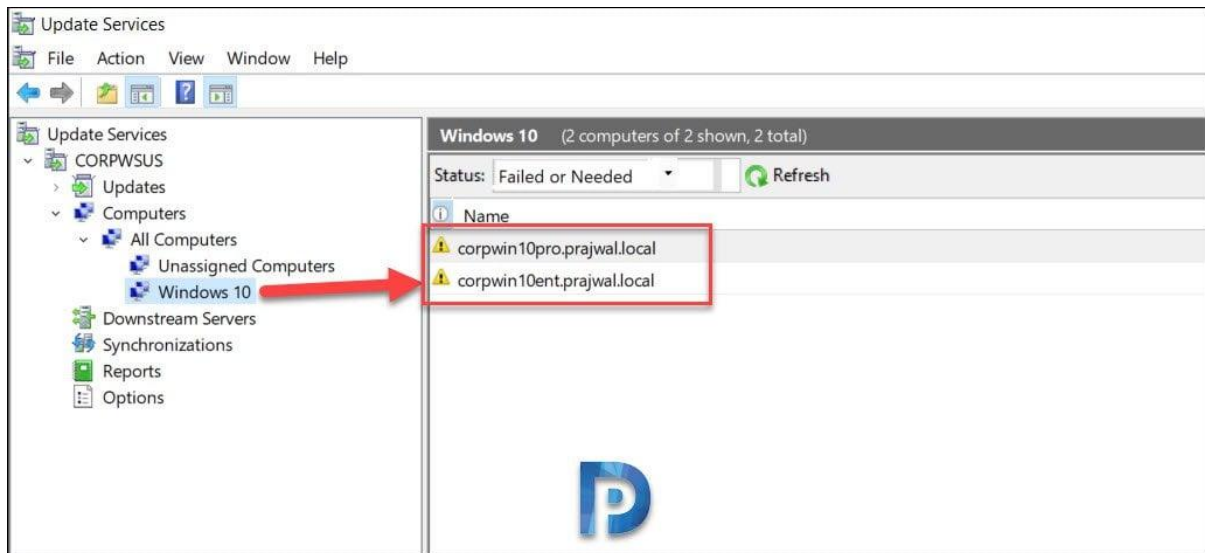
Change the computer membership

On the **Set Computer Group Membership** box, select the new group that you just created. Click **OK**.



Select target computer group

Click the new group and you should find those computers.

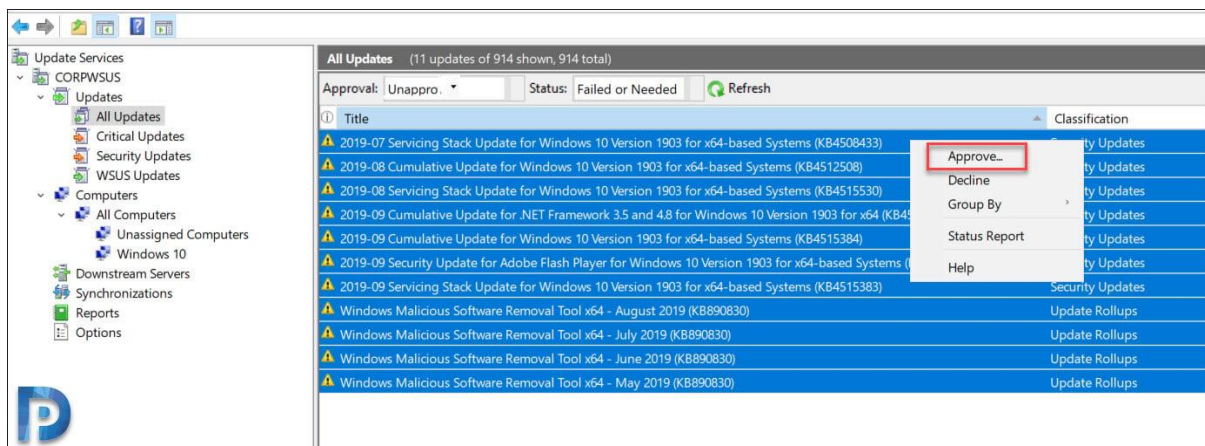


Approve and Deploy Updates in WSUS

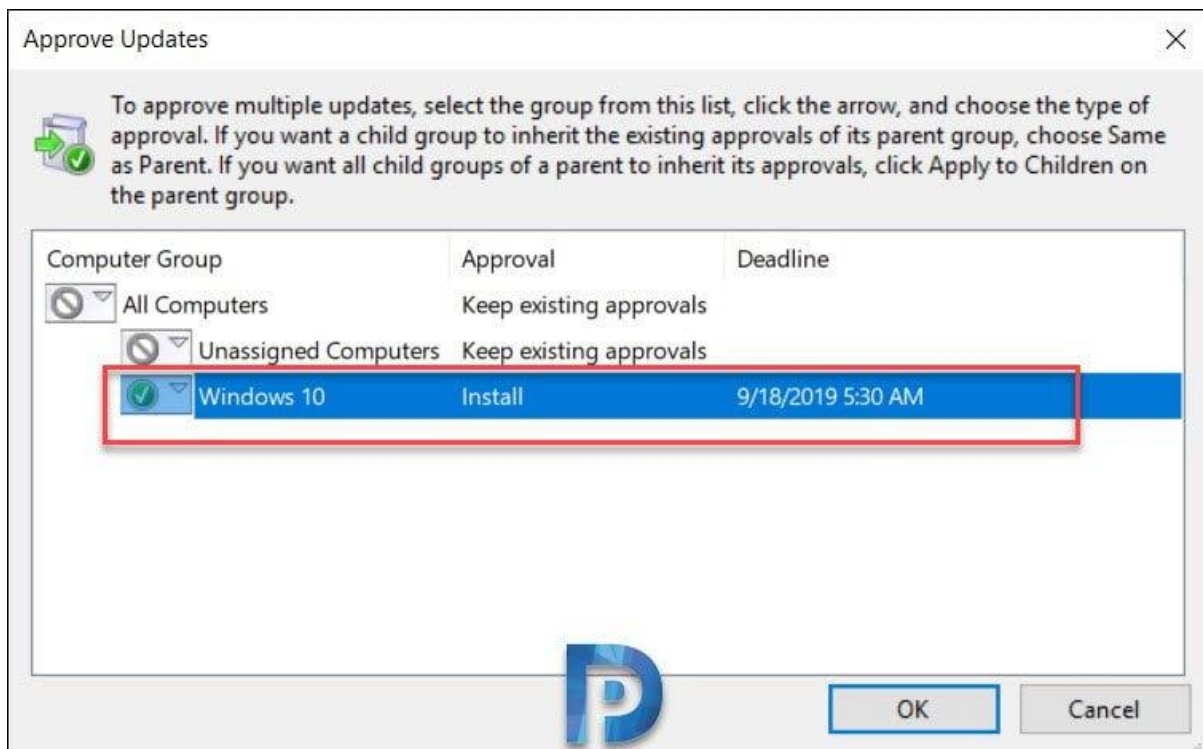
Once you have a test computer group created, your next task to deploy the updates to the test group. To do so you must first approve and deploy WSUS updates.

To approve the updates in WSUS

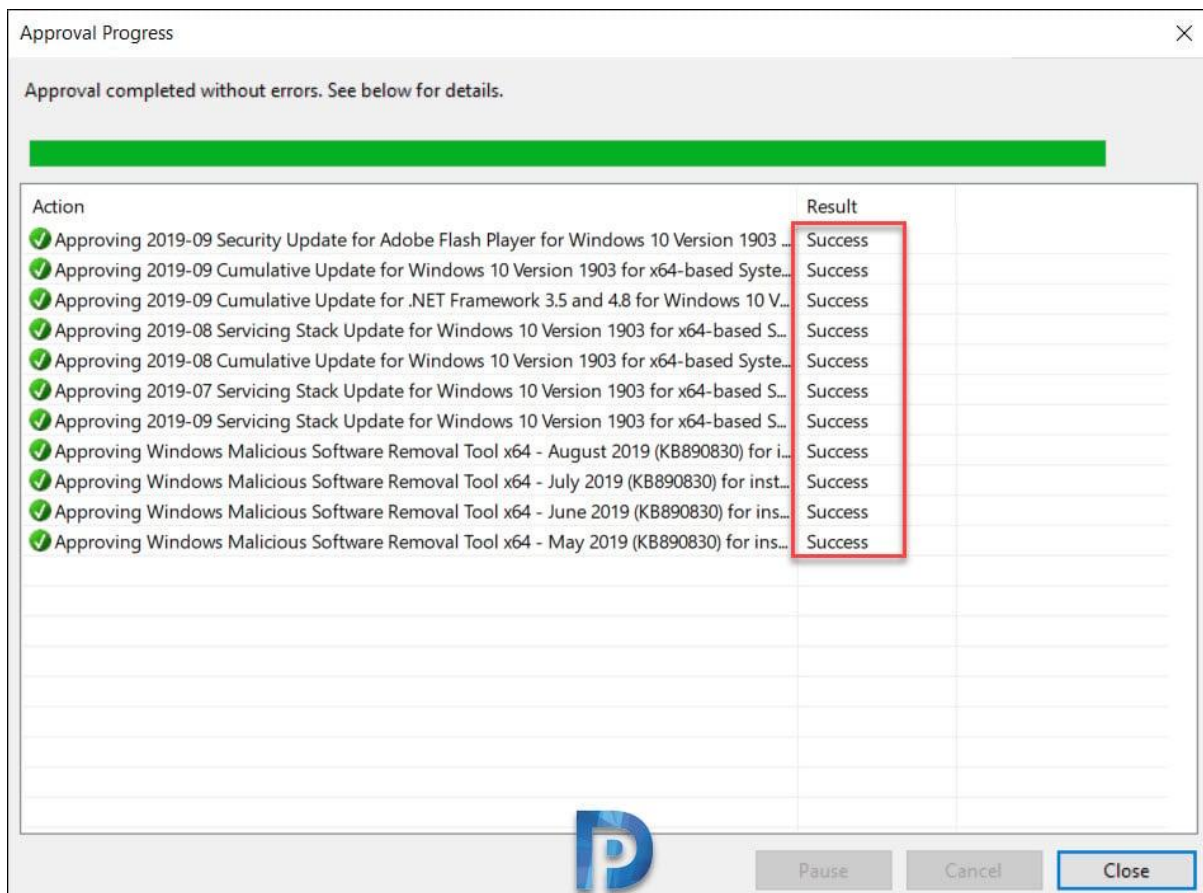
- Launch the WSUS Administration Console, click **Updates > All Updates**.
- In the **All Updates** section, select the updates that you want to approve for installation in your test computer group.
- Right-click the updates and click **Approve**.



Most of all in the **Approve Updates** dialog box, select your test group, and then click down arrow. Click **Approved for Install**. You can also set a deadline to install the updates. Click **OK**.



The Approval Progress window appears, which shows the progress of the tasks that affect update approval. When the approval process is complete, click **Close**.

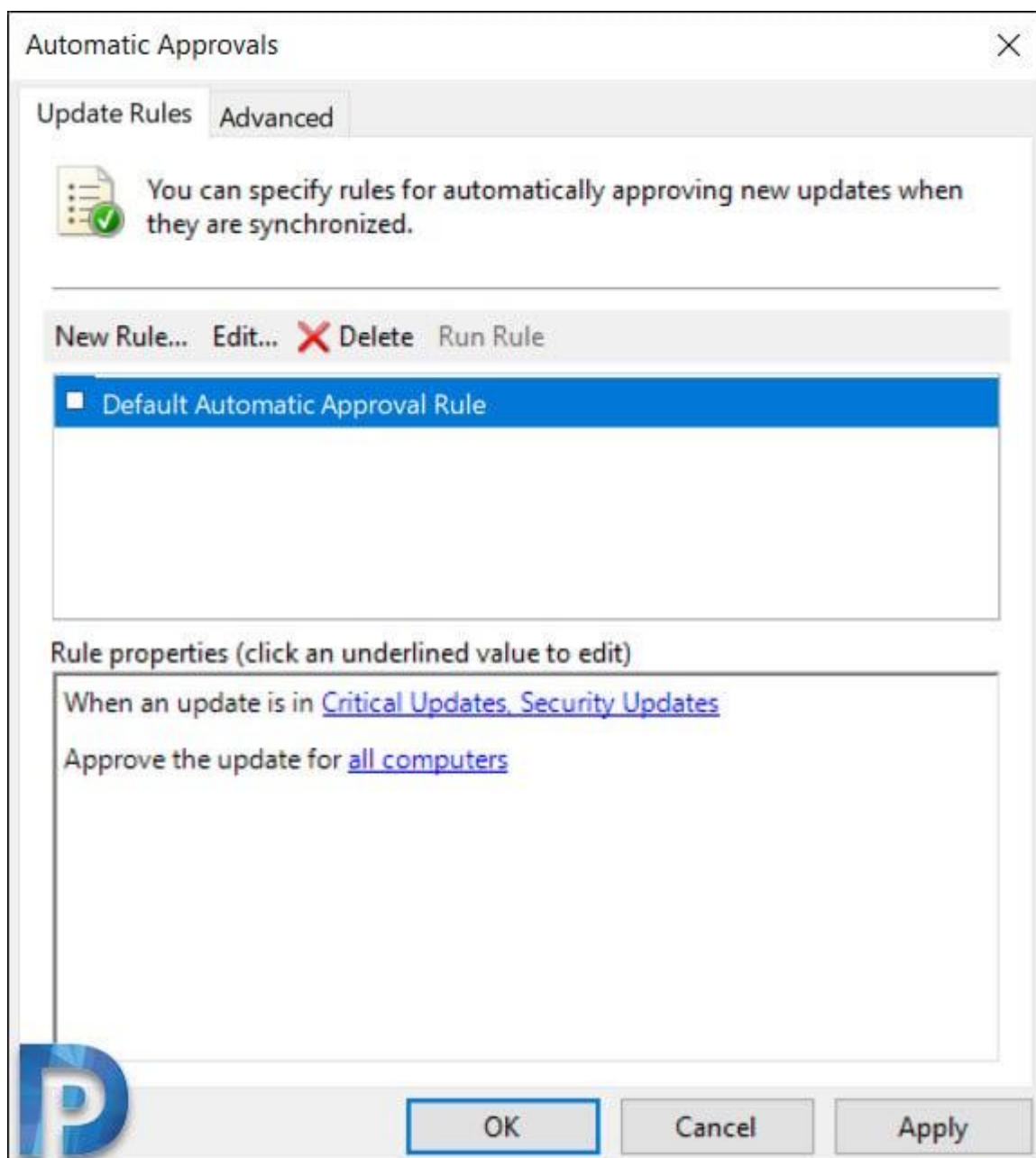


Configure Auto Approval Rules in WSUS

If you don't want to manually approve the updates you can configure auto approval rule in Windows Server Update Services.

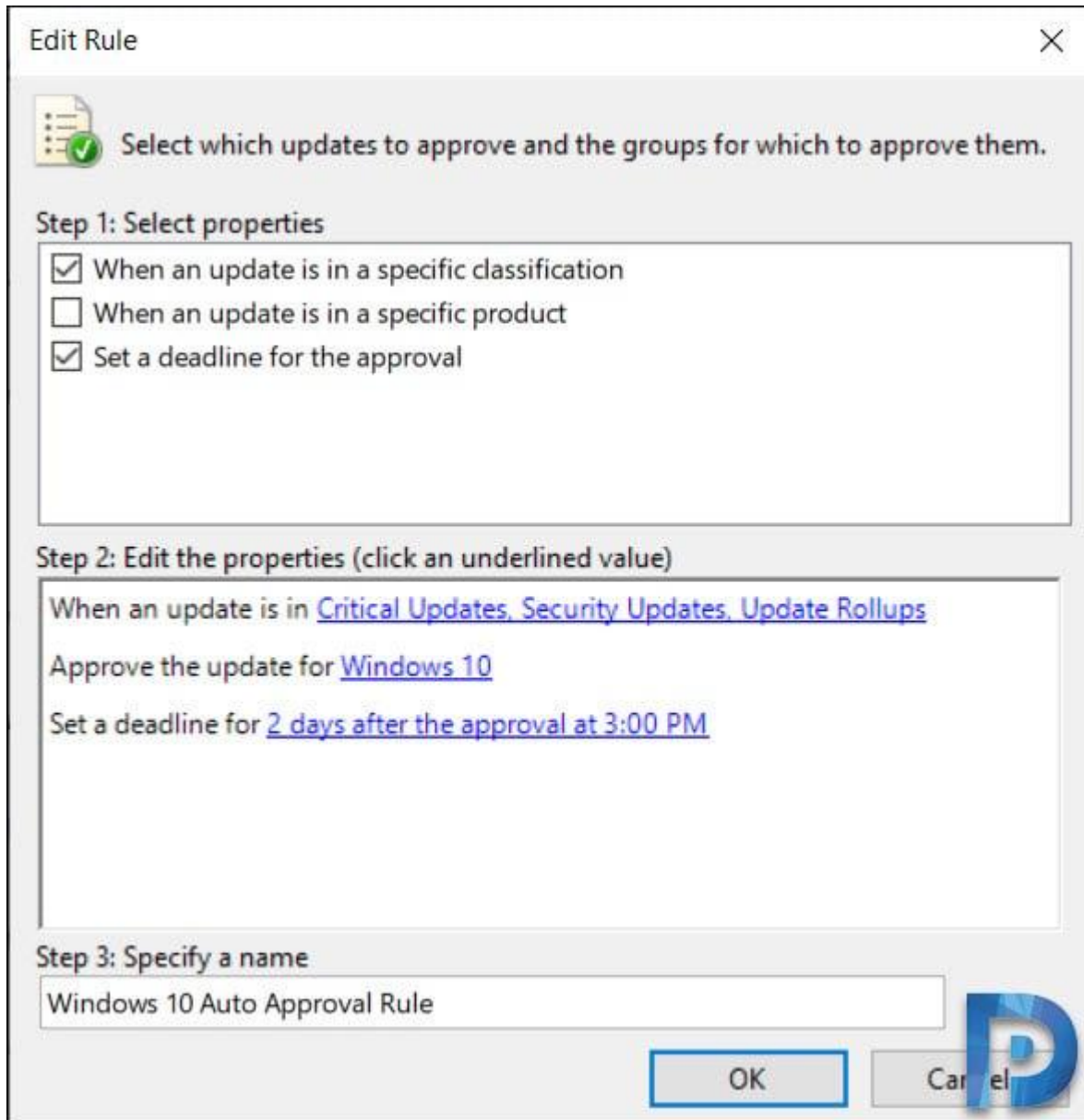
To configure Automatic Approvals in WSUS

- Launch WSUS Administration Console, expand the WSUS server, and then click **Options**.
- In Options, click **Automatic Approvals**.
- You should find the default automatic approval rule and if you wish you can edit it and use it.
- To create a new approval rule, click **New Rule**.



Check the box When an update is in a specific classification. Select the classifications. You can also approve the update for computers groups. I am going to select **Windows 10** as that is my test computer group. Finally you can set a deadline for the update approval and specify auto approval rule name.

After you configure the rule, click **OK**.



Edit Rule

Select which updates to approve and the groups for which to approve them.

Step 1: Select properties

- ☒ When an update is in a specific classification
- ☐ When an update is in a specific product
- ☒ Set a deadline for the approval

Step 2: Edit the properties (click an underlined value)

When an update is in Critical Updates, Security Updates, Update Rollups

Approve the update for Windows 10

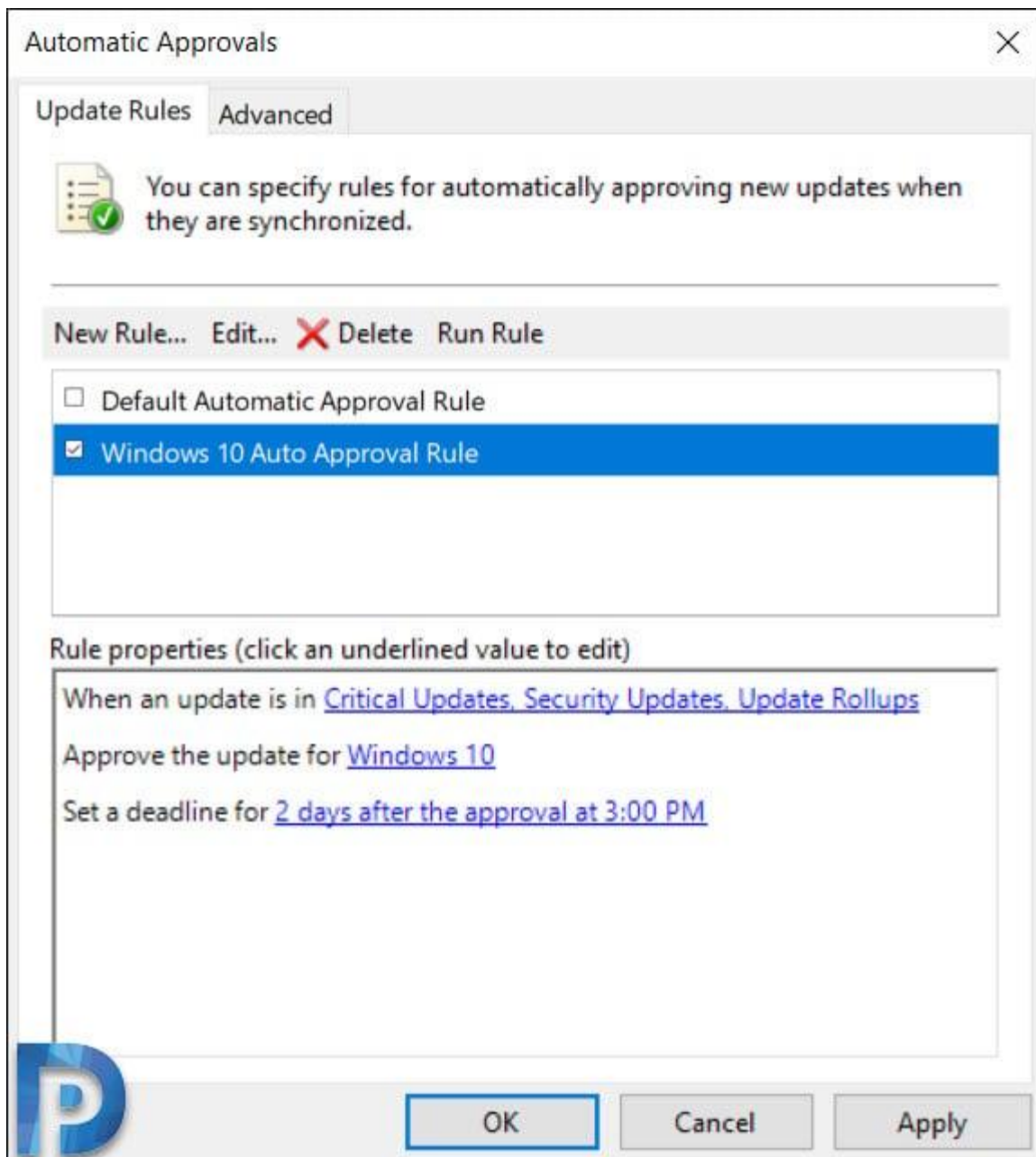
Set a deadline for 2 days after the approval at 3:00 PM

Step 3: Specify a name

Windows 10 Auto Approval Rule

OK Cancel

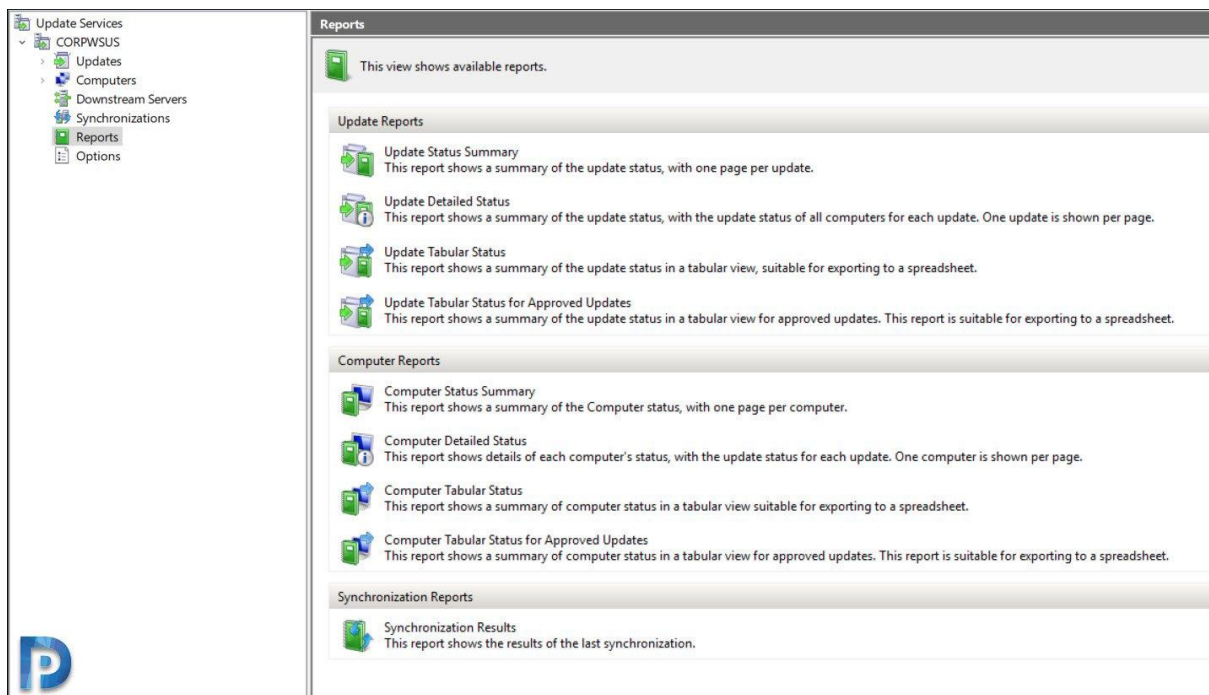
On the Automatic Approvals window, you can find the rule that you just created. If you wish to run this rule, click **Run Rule**.



WSUS Reports

The last section that I want to cover is the WSUS reports. Clicking **Reports** in the WSUS console shows the list of reports. WSUS comes with several reports to help you find the updates deployment status, sync reports and computers reports.

- **Update Reports** – Includes Updates status summary, detailed and tabular status, tabular status for Approved Updates.
- **Computer Reports** – Computer Status Summary, Detailed Status, Tabular Status and Computer tabular Status for approved updates.
- **Synchronization Reports** – Shows the results of last synchronization.



This completes the steps to install and configure WSUS